Local Income Taxes and Residential Sorting:

Evidence from Pennsylvania LEHD Origin-Destination Data

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Abstract

Do local income tax rate changes cause residential sorting across municipal boundaries? Using LEHD Origin-Destination data linking each worker's residence block to their workplace block at annual frequency, I estimate the effect of Pennsylvania municipal Earned Income Tax (EIT) rate changes on residential and workplace outcomes for ∼2,500 municipalities over 2002–2019. The PA EIT bill is piecewise: $T_{ji} = \max(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr,muni}})$, where j is the worker's home muni and i is the work muni. A rate hike only *moves the bill* when the relevant side of the max binds. I exploit this structure with a continuous stacked difference-in-differences design that decomposes each rate event by direction (Inc/Dec) and binding-state transition (already-binding, becomes-binding, non-binding-throughout), and that estimates the per-pp dose-response coefficient at the census block level—where both rate-setters' policy variation enters the regression at its native level—with municipality- and school-district-level aggregations reported as cross-checks. The within-design placebo is the Inc-nobind cell: rate hikes whose post-event home stack remains below the work-side nonresident rate, so the bill never moves and theory predicts no behavioral response. The headline finding is that a 1 percentage-point muni resident-rate increase in the

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binding cell reduces the block-level count of workers living in the block (in logs) by approximately 19 percent (-0.21 log points, SE $= 0.054$) with a clean realized-dose parallel-trends test ($\beta_{\text{pre}} = -0.027$, $|t| < 1$). The municipality-level analogue reduces the count of residents working in their own municipality (in logs) by approximately 33 percent (-0.40 log points) and raises the share of residents commuting out by 4.1 percentage points, though the muni-level pooled Inc-bind pre-trend fails parallel trends ($\beta_{\text{pre}} = +0.149$, $|t| = 2.66$), consistent with selection of post-industrial cities into rate hikes. The Inc-nobind placebo is null on the headline count outcomes at both levels. School district rate changes do not produce sorting at any binding state, even when SD changes flip the home-stack binding state. Industry-decomposed outcomes show consumer-facing jobs (retail, food, arts, other services) fall in the binding cell while tradeable jobs (manufacturing, wholesale, finance, professional services) are insulated, consistent with a local-services demand spillover from residents leaving rather than direct nonresident-tax incidence.

1 Introduction

The Tiebout hypothesis—that households "vote with their feet" by sorting across jurisdictions offering different bundles of taxes and public services (Tiebout, 1956)—remains a central organizing principle in local public finance. A large empirical literature documents tax-driven migration at the state level for high earners (Kleven et al., 2020; Young et al., 2016; **?**) and at sub-national borders for sales taxes (Agrawal, 2015). Yet sub-state income-tax sorting—the canonical Tiebout margin, where moving costs are small and tax boundaries are crossed by a short relocation—is essentially untested in the United States. Two recent studies of Indiana and Maryland local income taxes (**??**) use county-level IRS data and find county-level migration responses but cannot resolve the muni-level variation that defines the actual tax boundary. A literature survey by **?** explicitly identifies sub-state local-income-tax sorting as an open question, and **?** characterize the empirical evidence as "thin."

The gap persists for two reasons. First, the jurisdictions that levy local income taxes are sub-county, so standard migration data sources do not see them. The IRS Statistics of Income tracks county-to-county migration; the American Community Survey measures place-level population but does not link workers to employers. Second, local income-tax systems feature credit mechanisms, overlapping school-district taxes, and uneven enforcement that complicate the mapping from statutory rates to effective tax burden. A naive comparison of rates and population changes conflates tax incidence with amenity valuation, fiscal capacity, and enforcement quality.

This paper addresses both challenges using LEHD Origin-Destination Employment Statistics (LODES), the only administrative data source that links each worker's residence tract to their workplace tract at annual frequency. The setting is Pennsylvania, 2002–2019. Pennsylvania operates a universal Earned Income Tax under Act 511 of 1965 with continuous rate variation across approximately 2,500 municipalities and 500 school districts, and a quasi-experimental enforcement break: Act 32 of 2008 (effective January 1, 2012) consolidated EIT collection into 69 Tax Collection Districts and mandated employer withholding at the

higher of the home resident rate or the work nonresident rate. Before Act 32, withholding was fragmented and many municipalities did not even publish a nonresident rate. The structural break in enforcement—same statutory rates, very different effective tax burden—anchors causal identification.

This paper makes three contributions. First, it identifies the binding structure of the PA EIT as the central margin for empirical identification. The bill is piecewise in $\max(\tau^{\text{home}}, \tau^{\text{nr,work}})$, so a rate change only moves the bill when its side binds. I classify each muni-rate event by direction (Inc/Dec) and binding-state transition relative to the muni's nonresident rate, yielding seven cells (six event cells + NC controls). The within-design placebo is the Inc-nobind cell: rate hikes whose post-event home stack stays below the nonresident rate, so the bill never moves and theory predicts no behavioral response.

Second, the empirical design is a unified continuous stacked DiD with two simultaneous rate-setter treatments—muni resident rate and SD resident rate, each separately decomposed by binding cells—estimated at three units of observation (municipality, school district, census block). Triangulation across estimating units replaces the cross-estimator triangulation of an earlier draft: each level aligns the regression with a different rate-setter's jurisdiction and exposes different pre-trend behavior. Pre-trend tests are reported via two complementary diagnostics: a realized-dose Sun-Abraham continuous event study (**?**de Chaisemartin et al., 2024) and a binary Sun-Abraham IW event study (Sun and Abraham, 2021). Controls are share-based throughout: police, streets, and debt service as shares of total expenditures; EIT, real estate, and intergovernmental revenue as shares of total revenues. The never-changed control pool is strictly defined as munis whose rate never changed across the panel, not the broader set of munis whose first event could be cell-classified.

Third, the headline finding is that the muni resident rate moves residents through the binding channel only. The block-level estimate (Inc-bind on the log count of workers living in the block) is -0.21 log points per pp—approximately a 19 percent decline in resident workers per percentage-point of $\tau^{\text{res,muni}}$, with a clean realized- dose parallel-trends diagnostic. The

municipality-level analogue is larger (-0.40 log points on the log count of residents working in their own municipality, $\approx 33\%$ per pp; -0.27 on log total jobs, $\approx 24\%$; -0.10 on log employed residents, $\approx 10\%$; and the share of residents commuting out rises 4.1 percentage points) but its pre-trend diagnostic fails parallel trends, consistent with selection into rate hikes by declining post-industrial cities. The Inc-nobind placebo cell—rate hikes whose post-event home stack stays below the nonresident rate—is statistically null on the headline count outcomes at both levels. School district rate changes do not produce sorting in any cell, even when SD changes flip the binding state, consistent with workers responding to the bill rather than to the rate's identity. Industry decomposition using LODES WAC at the municipality level shows consumer-facing jobs (retail, food, arts, other services) fall in the binding cell while tradeable jobs (manufacturing, wholesale, finance, professional services) are insulated, consistent with a local-services demand spillover from residents leaving rather than direct nonresident-tax incidence.

The municipality-level magnitude is concentrated in 8 post-industrial PA cities (Wilkes-Barre, Hazleton, Johnstown, Jeannette, Coatesville, Warren, Carbondale, Farrell); the 32 borough and township Inc-bind munis show direction-consistent but individually insignificant effects. The pre-trend failure at the municipality level is attributable to this composition: high-eventual-dose cities are on declining trajectories before their rate changes. The block-level specification absorbs the gentrification-correlated pre-trend at finer geography, which is why I report the block-level estimate as the headline causal quantity and the municipality-level estimate as the typical-jurisdiction magnitude calibration.

2 Institutional Background

Pennsylvania's Earned Income Tax (EIT) is the largest source of local-income-tax variation in the United States. Approximately 2,500 municipalities and 500 school districts each set their own rate, covering essentially all earned income earned in or by Pennsylvania

residents. Unlike property taxes, which are paid by owners and capitalized into home values, the EIT falls directly on workers and varies across municipal boundaries within the same metropolitan area. This creates sharp spatial discontinuities in tax burden that can influence where workers choose to live and work.

2.1 Pennsylvania Earned Income Tax

Pennsylvania's EIT is mandatory for all municipalities under the Local Tax Enabling Act (Act 511 of 1965). It applies to all earned income with no minimum threshold; retirees are exempt. The standard combined cap is 1%, typically split as 0.5% municipal and 0.5% school district. Home Rule municipalities may exceed this cap, and Philadelphia operates separately under the Sterling Act of 1932 at approximately 3.9%. The EIT is the primary revenue lever for most Pennsylvania municipalities: property tax millage caps are binding in many jurisdictions. Our panel (2002–2019) contains 210 muni rate increases, 119 muni rate decreases, 611 school-district rate increases, and 52 school-district rate decreases.

Three rates per municipality. Each Pennsylvania municipality has three EIT rates relevant to a worker:

- $\tau^{\text{res,muni}}$, the muni resident rate, levied on residents of the muni;
- $\tau^{\text{nr,muni}}$, the muni nonresident rate, levied on workers employed in the muni who live elsewhere;
- τ^{SD}, the school-district rate, levied on residents only (school districts cannot tax non-residents).

Each address is associated with a six-digit Political Subdivision (PSD) code identifying its (muni, school district) pair. Workers report this code on the Residency Certification Form, and employers use it to determine withholding obligations.

The piecewise tax allocation. Under Act 32 (effective January 1, 2012), a worker's total EIT bill is determined by the maximum of the home-side rate stack and the work-location nonresident rate, and the bill is split piecewise between home and work jurisdictions:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr,muni}},\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr,muni}} \\ \left(0,\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr,muni}} \end{cases} \tag{1}$$

where j is the worker's home muni and i is the work muni. The work jurisdiction always collects its full nonresident rate first; the home muni and school district then split whatever portion of the worker's bill is left over after the work claim. If the work nonresident rate exceeds the home stack, the home jurisdictions get nothing. The worker's total bill is the higher of the two stacks:

$$\text{Bill}_{ji} = \max\left(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr,muni}}\right). \tag{2}$$

Implications for sorting. Equation (1) implies which margin a worker can use to respond to a rate increase. When the home stack binds ($\tau_j^{\text{home}} > \tau_i^{\text{nr,muni}}$), the worker can lower the bill only by moving home to a lower-rate muni or school district. When the work nonresident rate binds, the worker can lower the bill only by changing job to a lower-nonres-rate work location; moving home without changing job has no effect. For workers who live and work in the same muni, only the home stack applies—they pay $\tau_i^{\text{res,muni}} + \tau_i^{\text{SD}}$.

A worker who lives in muni j and moves to a lower-rate muni j' while keeping the same job at i saves at least the SD rate at j: the new bill is $\max(\tau_{j'}^{\text{home}}, \tau_i^{\text{nr}})$, which is at most the old bill τ_j^{home} minus τ_j^{SD}. The often-stated claim that "moving residence under a credit system yields zero tax savings" is wrong under Pennsylvania's piecewise structure—residence moves do save tax, because the home-side stack includes the SD rate that does not reappear at the work side.

Philadelphia (excluded from the Act 32 framework). Philadelphia operates under the Sterling Act of 1932 rather than Act 32. Philadelphia residents pay the city resident rate regardless of work location with no offset; Philadelphia nonresidents pay the city nonresident rate, with no residual flowing to home jurisdictions in other Pennsylvania munis. Our main estimates retain Philadelphia in the sample for completeness; results are robust to dropping it.

Act 32: Tax Collection Consolidation. Before 2012, EIT collection was radically decentralized: each of Pennsylvania's 2,500+ municipalities appointed its own tax officer. Nonresident EIT enforcement was weak because no centralized system linked employers, workers, and municipalities. A worker who moved out of a rate-raising municipality could, in practice, avoid paying the nonresident portion to the work municipality even though they legally owed it. The piecewise rule described in Equation 1 existed on paper but was not effectively enforced.

Act 32 of 2008 (effective January 1, 2012) transformed this system by consolidating collection into approximately 67 county-based Tax Collection Districts (TCDs), each with a single appointed collector. The law mandated standardized employer withholding at the higher of the employee's resident and work rate. This had two consequences. First, it made the EIT *binding*: rate changes now mechanically appear in every worker's paycheck through employer withholding, regardless of whether they reside in the taxing municipality. Second, it eliminated the possibility of de facto tax avoidance through residential relocation. Before Act 32, moving away from a rate-raising city could effectively reduce a worker's tax burden because the work municipality lacked the enforcement capacity to collect from nonresidents. After Act 32, the credit system is fully enforced, and moving away while keeping the same job provides no tax relief.

School District EIT. School districts in Pennsylvania also levy an EIT under Act 511, typically at 0.5% (the other half of the standard 1% cap). Act 1 of 2006 (the Taxpayer

Relief Act) additionally authorized school districts to exceed the 0.5% cap specifically to fund property tax relief. This generated a wave of school district rate increases in 2006–2011, creating rate variation that is largely independent of municipal fiscal decisions: the correlation between municipal and school district EIT rates across our panel is -0.12. Since every census tract sits in both a municipality and a school district, workers face two overlapping EIT rates that can change independently, providing a natural decomposition of the residential sorting response into municipal and school district components.

3 Conceptual Framework: Binding-State Decomposition

The piecewise tax allocation in Equation (1) is the analytical engine of the empirical design. A worker's bill is $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$. A rate change only *moves the bill* when its side of the max binds. This generates a sharp prediction: the same statutory rate change can produce a strong behavioral response, a moderate response, or no response at all depending on the binding state pre- and post-event.

3.1 Six Cells per Side

For each muni i and each rate variable $\tau \in \{\tau^{\text{res,muni}}, \tau^{\text{SD}}, \tau^{\text{nr,muni}}\}$, classify the muni's first observed rate event by direction (Increase / Decrease) and binding-state transition relative to the muni's own home-stack vs nonresident rate comparison:

- **Inc-bind**: rate up; home stack was binding pre and post. Bill rises by full $\Delta\tau$ for in-muni residents. *Predicted: strongest behavioral response.*
- **Inc-becomes**: rate up; home stack flips from non-binding to binding. Bill rises by less than full $\Delta\tau$ (only the portion above the work nonres rate). *Predicted: intermediate response.*

- **Inc-nobind**: rate up; home stack stays below nonres rate throughout. *Bill never moves. Predicted: no response. This is the within-design placebo.*

- **Dec-bind**: rate down; home stack stays binding. Bill falls by full $\Delta\tau$. Predicted mirror of Inc-bind under symmetry.

- **Dec-becomes-non**: rate down; home stack flips from binding to non-binding. Bill falls only down to nonres rate.

- **Dec-nobind**: rate down; home stack stays below nonres throughout. Bill never moves. Second placebo cell.

A seventh category, NC (*never-changed*), comprises munis whose τ never changes across the panel. NC serves as the control pool.

The fundamental testable prediction of the institutional rule is that **Inc-bind effects > Inc-becomes effects > Inc-nobind effects ≈ 0 in absolute value**, with Inc-nobind serving as the within-design placebo. If the Inc-nobind cell shows behavioral response, the design's mechanism interpretation fails—the response is something other than the binding rule.

3.2 Three Treatment Margins

PA's institutional structure separates three rate variables that change independently across muni-years:

- $\tau^{\text{res,muni}}$ (muni resident rate) directly affects the bill of in-muni residents through the home-stack channel.

- τ^{SD} (school district rate) also affects the home stack but is set by school districts, not municipalities. Predicted: same binding-rule mechanism as muni rate, but the SD's institutional decision-making is different from the muni's.

- $\tau^{\text{nr,muni}}$ (muni nonresident rate) affects nonresident workers commuting in. The piecewise rule means it moves the bill of workers from low-home-stack origins.

The two predictions across treatments:

1. Each rate's binding-cell decomposition shows the same Inc-bind > Inc-becomes > Inc-nobind pattern.

2. Channel separation: muni resident rate moves residents (live-and-work, total residents); muni nonresident rate moves nonresident workers (in-commuters); SD rate's effect depends on whether SD changes induce similar residential responses.

4 Data

4.1 Data Construction Pipeline

The analysis links two administrative systems that do not share common identifiers: LEHD/LODES employment records (Census block geography) and Pennsylvania EIT rates (DCED municipality IDs). The build sequence scrapes the DCED MunStats portal for the rate panel, scrapes the DCED Annual Financial Reports for muni fiscal controls, downloads block-level LODES files for 2002–2019 (a ∼5 GB raw OD database), crosswalks PA municipalities to TIGER COUSUBFP codes by normalized name and county (99.6% match), builds a block-to-municipality crosswalk by point-in-polygon assignment of block centroids to 2020 cousub geometries (Census blocks nest in cousubs by construction), and aggregates block-pair-year LODES flows to the municipality, school-district, and block levels of observation. Step-by-step pipeline detail (scraper specs, crosswalks, imputation rules, and the auxiliary LODES file for cross-state inflows) is in the Companion Appendix (`appendix.tex`).

4.2 Data Sources

LODES Origin-Destination Data. LEHD LODES Version 8, 2002–2019. Block-level residence–workplace links derived from state Unemployment Insurance wage records and the Quarterly Census of Employment and Wages (**?**). We trim 2020–2021 due to COVID-related disruptions in both employment patterns and LODES coverage. The Census Bureau applies synthetic noise infusion at the block level to protect respondent confidentiality; the mechanism is calibrated so that aggregation to larger geographies produces estimates that approximate ground truth and have been validated against external sources at the tract level and above (**???**). We aggregate blocks to county subdivisions (municipalities) using the TIGER 2020 block-to-municipality crosswalk described in Section 4.1, item 3.

Municipal Tax Rate Panel. Pennsylvania: DCED MunStats, scraped annually. Resident and nonresident municipal EIT rates, school district EIT rates for all ∼2,500 municipalities. Name-based crosswalk to Census COUSUBFP (99.6% match rate). Pre-2012 nonresident rates backfilled from earliest observed post-2012 value per muni (the registry was incomplete pre-Act 32).

Municipal Financial Reports. PA DCED Annual Financial Reports, 2002–2019 (trimmed from 2002–2023 to exclude COVID/ARPA fiscal distortion). 71 columns per municipality-year including EIT revenue, real estate tax revenue, police expenditure, streets expenditure, total debt, state intergovernmental aid, population, and assessed market value of real estate.

Crime. FBI Uniform Crime Reports, Offenses Known and Clearances by Arrest (Jacob Kaplan concatenated files), 2002–2019. Monthly agency-level counts of Part I index crimes (murder, robbery, aggravated assault, burglary, larceny-theft, motor vehicle theft). Aggregated to annual totals per agency. Matched to PA municipalities via `fips_place_code` (93.5% agency match rate).

Building Permits. Census Bureau Building Permits Survey (BPS), place-level annual files, 2002–2019. Total housing units permitted by structure type (1-unit, 2-unit, 3–4 unit, 5+ unit). PA municipalities matched via `fips_mcd_code` = COUSUBFP (93.3% MCD match rate).

4.3 Sample Restrictions and Transformations

1. **Sample period: 2002–2019.** LODES coverage begins in 2002. We trim at 2019 because COVID disrupted both commuting patterns (LODES 2020–2021) and municipal finances (ARPA transfers distort AFR data 2020–2023). Including these years would conflate pandemic-driven relocation with tax-motivated sorting.

2. **Minimum employment threshold: 50 total jobs.** Small municipalities have share_live_in values dominated by sampling noise (a single worker moving changes the ratio by $1/N$). The 50-job threshold balances precision against sample size. Sensitivity analysis (Section 7) confirms stability across thresholds from 10 to 500.

3. **Rate outlier filters.** PA DCED reports the Local Services Tax (a flat dollar amount) in the same field as the EIT rate (a percentage) in some years. Dropping rates $>$ 5% removes these data entry errors without affecting any real EIT rate (the highest standard EIT is $\sim$1%; only Philadelphia exceeds 3%).

4. **April 1 alignment.** LODES records employment as of April 1. A rate change that takes effect mid-year could be misattributed if not aligned. We assign each municipality-year the rate in effect on April 1, ensuring the tax rate a worker faces and the OD flow recording their location refer to the same date.

5. **Outcome variables.** The primary outcome is $\textit{share_live_in}_{it} = \text{live_and_work}_{it}/\text{total_jobs}_{it}$, the locally-resident fraction of a city's workforce. Unlike population counts (which mix retirees, students, and non-workers) or IRS migration data (which is county-level), share_live_in isolates the residential choices of active workers at municipal resolution.

To decompose the share response into levels, we also estimate effects on log(resident workers) and log(total jobs) separately. Fiscal outcomes include log EIT revenue, log total tax revenue, log total revenue, log police expenditure, log population, and log market value of real estate from the PA DCED Annual Financial Reports. Pre-treatment balance tests cover eleven observables including crime and building permits.

6. **Treatment definitions.** We use both a continuous and a binary treatment definition. *Continuous*: the current rate level (resident municipal, school district, and the home stack as defined in Equation 1), capturing the intensive margin via TWFE and DCDH. *Binary cohorts*: 1 after a municipality's first rate increase (or first decrease), enabling Sun-Abraham estimation of staggered-DiD-robust dynamic effects.

7. **Control group.** Pennsylvania's EIT is universal, so the relevant control group is municipalities with a stable rate over 2002–2019 (1,264 never-changed municipalities). For the Q3 clean-groups specification we further partition treated municipalities into increase-only (97) and decrease-only (24), dropping the flip-floppers that change in both directions.

8. **Fiscal controls.** Rate increases may coincide with other fiscal changes (police hiring, debt issuance) that independently affect residential attractiveness. We include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid as time-varying controls, plus revenue shares (EIT, real-estate, intergovernmental) from the PA DCED AFR. The main estimates are unchanged with and without these controls (the robustness section (Section 7)).

Table 1: Summary Statistics

Variable	Mean	Median	SD	Min	Max
share_live_in	0.149	0.127	0.097	0	0.700
Resident EIT rate (%)	0.542	0.500	0.199	0	4.331
Total jobs	3,624	1,224			
Resident workers	674	154			
Municipality × years	26,650 (1,479 municipalities)				

Notes: Pennsylvania municipality-year panel, 2002–2019. Sample restricted to municipalities with > 50 total jobs. share_live_in = workers who live and work in municipality / total workers in municipality. Municipalities are Census county subdivisions.

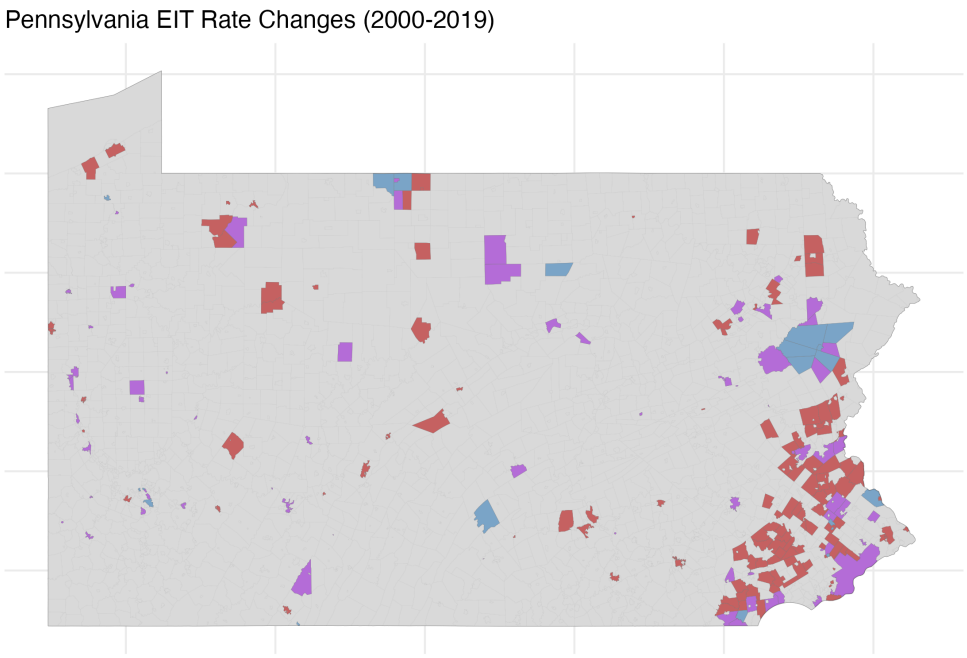


Figure 1: Pennsylvania EIT Rate Changes (2000–2019). Red = increased; blue = decreased; purple = both; grey = no change.

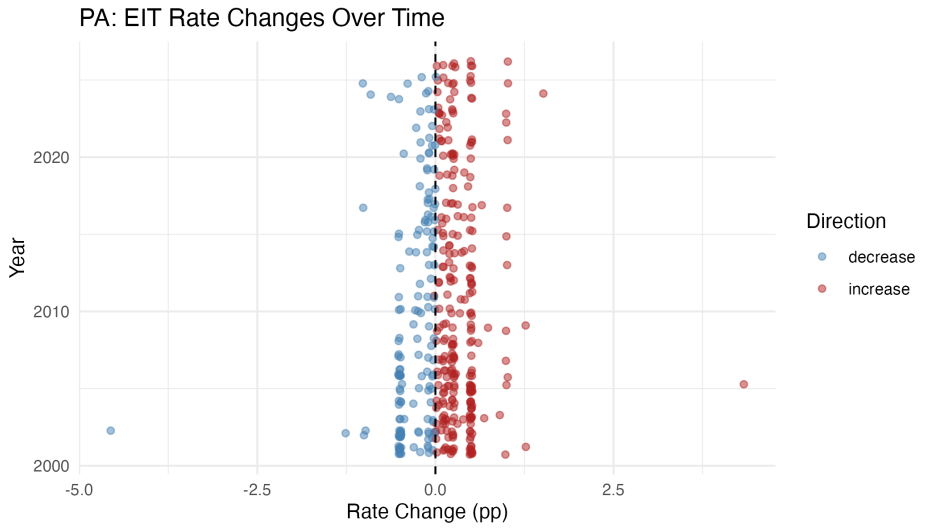


Figure 2: Pennsylvania: Distribution of EIT Rate Changes Over Time

5 Empirical Strategy

The empirical framework is a continuous stacked difference-in-differences design with binding-state cell decomposition, estimated at three units of observation: municipality, school district, and census block. Each estimating unit aligns the regression with a different level of treatment aggregation, producing a triangulation across units rather than across estimators. The first-difference dose-response specification estimates the post-event effect of a one percentage-point change in the relevant tax rate; a separate realized-dose event-study diagnostic tests for differential pre-event trajectories across units of varying eventual treatment intensity.

5.1 Setup

This subsection describes the elements of the design that are common across the three estimating units (municipality, school district, census block); unit-specific choices for treatment construction, fiscal controls, fixed effects, and clustering are described in Section 5.5.

Sample period. 2002–2019. Pre-2002 data are not consistently available in LODES;

post-2019 years are excluded because COVID-19 disrupted commuting patterns and ARPA fiscal transfers distorted muni budgets.

Outcomes. The continuous-stacked-DiD specification is run in parallel across a set of municipality-level workplace and residence outcomes: log of the count of `live_and_work` (residents employed in the muni), `in_commuters` (workers in the muni from elsewhere), `out_commuters` (residents employed elsewhere), `total_jobs`, and `total_residents`; plus the shares $the share of workers living in their own muni$, $the share of workers who are nonresidents$, and $the share of residents working outside the muni$. SD-level regressions use the same outcomes aggregated to the SD via the muni–SD assignment. At the block level, outcomes are redefined as block-level analogues (the log count of workers living in the block, the log count of workers at the block, the share of block residents working in their own muni, the share of residents working outside the muni); see Section 5.5.

Two rate treatments. The unified resident-side specification includes two simultaneous rate-setter treatments: $\Delta\tau_{it}^{\text{muni}}$ (cumulative change in the muni resident rate from baseline) and $\Delta\tau_{it}^{\text{SD}}$ (cumulative change in the school district rate). Baseline for each is the rate at year $c_i^s - 1$ where c_i^s is the unit's first event year on side $s \in \{\text{muni, SD}\}$. Cumulative dose accumulates with subsequent rate changes within unit i. The work-side nonresident rate τ^{nr} enters the binding-state classification of cells (Section 5.2) but is not itself a treatment in the headline regression; we focus on the resident-side mechanism in this paper and defer the work-side behavioral response to a separate analysis.

Cell decomposition. Each rate-change event is classified by direction $\in \{\text{Inc, Dec}\}$ and binding-state transition $\in \{\text{bind, becomes, nobind}\}$, yielding six non-NC cells per side. The classification rule is given in Section 5.2. Units whose rate τ^s never changes across the panel are assigned to the NC reference category on side s.

Fiscal controls. Share-based, all time-varying. The exact control set differs by estimating unit and is described in Section 5.5. Municipality-level regressions use muni AFR shares (police, streets, debt service / total expenditures; EIT, real estate, intergovernmen-

tal / total revenues). SD-level regressions use Census F-33 SD shares (state, local, federal, property-tax revenue shares; instructional, support, capital, debt- interest expenditure shares). Block-level regressions use both, since each block is exposed to both rate-setters' fiscal posture.

5.2 Cell Classification (per side)

The six binding-state cells are defined in Section 3.1. For each unit i and each rate-setter side $s \in \{\text{muni, SD}\}$, I identify the first rate event year c_i^s and assign the unit to one of the six non-NC cells based on the direction of the change and the binding state of the home stack relative to $\tau^{\text{nr,muni}}$ in the pre- and post-event years. Units whose rate τ^s never changes across 2002–2019 are assigned to the NC reference category on side s.

NC purity. The NC category is restricted strictly to units whose rate is constant over the full panel. Units with rate changes whose first event has an undefined binding state—typically because the muni's nonresident rate is missing in the pre-event year due to DCED reporting gaps before Act 32—are categorized as "Uncertain" and dropped from the analysis. Pooling them with NC would contaminate the control trend with treated-but-unclassified units; reporting them in their own cell would mix true binding states with measurement gaps.

5.3 Headline Estimator: Stacked Continuous DiD

The headline estimator is stacked continuous difference-in-differences. For each side s and each cohort year c at which any muni's first event on side s occurs, build a sub-experiment $\mathcal{S}_c^s$:

- **Treated set:** munis i with $c_i^s = c$ and $\text{cell}_i^s \neq \text{NC}$.
- **Control set:** all NC^s munis (never-changed on side s).
- **Window:** years $t \in [c - K, c + K]$ where $K = 5$.

- **Stack indicator:** $s_i = c$ for treated muni i; $s_i = c$ for each NC muni replicated in this stack.
- **Relative-event time:** $\ell_{it} = t - c$.
- **Stack-muni and stack-year identifiers:** $\tilde{i} = (s_i, i)$, $\tilde{t} = (s_i, t)$.

Stacking all sub-experiments yields a single panel where each NC muni appears once per cohort and each treated muni appears in exactly one sub-experiment. Within each sub-experiment, controls are clean (no treated muni from another cohort serves as control), avoiding the negative-weighting problem in vanilla TWFE under staggered treatment (**?**).

The stacked-DiD construction is in the lineage of **?** (138 minimum-wage events, stacked event-by-event with continuous treatment), **?** (severe health shocks, stacked with future-treated households as controls), and **?** (SSA field-office closings, stacked with future-closing ZIP codes as controls). Three differences are worth flagging.

Control-group choice. **?** and **?** use *future-treated* units (those that will be treated at $\tau + \Delta$) as controls, because never-treated units are scarce or selected in their settings (no never-dying households; selection into closing eligibility). **?** uses *clean controls* (states with no minimum-wage events in the 8-year window around the focal event), which permits states with events outside the window to serve as controls. We use the strictest specification: *never-changed* (NC) munis, defined as those whose rate τ^s never changes across 2002–2019. Since 95% of PA municipalities are NC on at least one rate side, the never-changed pool is abundant and we do not need to relax to future-treated or clean-within-window controls.

Continuous treatment. **?** is the closest analogue on this dimension: their treatment is continuous (log minimum wage) within each stacked sub-experiment. Their identification logic—"aligning events by event-time, not calendar time, equivalent to a setting where the events happen all at once" (**?**, Online Appendix C)—applies identically in our setup. We extend their continuous-treatment design by extracting *cell-specific* dose-response coefficients via the $i(\text{cell}, \Delta\tau)$ interaction, which their pooled event-by-event spec does not produce.

Cell decomposition. The binding-state cell partitioning is new relative to all three. **?** pools all minimum-wage events without a within-design taxonomy of treated units; **?** and **?** have homogeneous treatments (death, closure). Our decomposition exploits the piecewise PA EIT rule to partition treated munis into cells whose ex ante predicted response differs by binding state, generating a within-design placebo (Inc-nobind) that the standard stacked DiD cannot produce.

Multi-event treated munis. A treated muni i may have more than one rate-change event over 2002–2019. We classify each such muni by the binding-state transition of its *first* event and assign i to the corresponding cell on side s. The muni then appears in exactly one sub-experiment (its first-event cohort c_i^s), not in subsequent cohorts triggered by later events.

Subsequent rate changes within the same muni accumulate into $\Delta\tau_{it}^s$, the *cumulative* dose from the muni's pre-event baseline. Concretely: suppose muni i has $\tau_i^s = 0.50$ in years 2005–2009, $\tau_i^s = 0.70$ in 2010 (first event, $c_i^s = 2010$, baseline = 0.50), and $\tau_i^s = 1.00$ from 2014 onward (second event). Then $\Delta\tau_{it}^s = 0$ for $t < 2010$, $\Delta\tau_{it}^s = 0.20$ for $t \in [2010, 2013]$, and $\Delta\tau_{it}^s = 0.50$ for $t \geq 2014$. The stacked-continuous specification absorbs this trajectory through the dose-stack interaction in a single regression on muni i's 2005–2015 window (its first-event cohort 2010 with $K = 5$). Muni i is *not* duplicated as a 2014 cohort treated muni; it stays in its 2010 cohort. The cumulative dose path encodes the second event by raising $\Delta\tau_{it}^s$ from 0.20 to 0.50 at year 2014.

This first-event classification has three deliberate consequences. First, no treated muni serves as control to itself; each muni appears in exactly one stack. Second, multi-event treated munis do not receive disproportionate weight from appearing in multiple stacks. Third, the per-pp coefficient β_k^s is interpretable as a dose-response elasticity over the cumulative rate change, which is the welfare-relevant magnitude for fiscal calibration.

For each treated muni i in stack $\mathcal{S}_c^s$, define the cumulative dose-stack as $\Delta\tau_{it}^s = \tau_{it}^s - \tau_{i,c-1}^s$,

with $\Delta\tau^s_{it} = 0$ for $t < c$ (pre-event) and for NC munis at all t. The stacked continuous DiD specification, for outcome Y, runs the regression on the stacked panel:

$$\begin{aligned} Y_{i\tilde{t}} \;=\; \alpha_{\tilde{i}} \;+\; \gamma_{\tilde{t}} &+ \sum_{k \in \mathcal{C}_{\setminus \mathrm{NC}}} \beta^{\mathrm{muni}}_k \, \Delta\tau^{\mathrm{muni}}_{it} \cdot \mathbf{1}\{\mathrm{cell}^{\mathrm{muni}}_i = k\} \\ &+ \sum_{k \in \mathcal{C}_{\setminus \mathrm{NC}}} \beta^{\mathrm{SD}}_k \, \Delta\tau^{\mathrm{SD}}_{it} \cdot \mathbf{1}\{\mathrm{cell}^{\mathrm{SD}}_i = k\} \\ &+ \mathbf{X}'_{it}\boldsymbol{\phi} \;+\; \varepsilon_{i\tilde{t}}, \end{aligned} \tag{3}$$

where

- $\mathcal{C}_{\setminus \mathrm{NC}} = \{$Inc-bind, Inc-becomes, Inc-nobind, Dec-bind, Dec-becomes-non, Dec-nobind$\}$ is the set of non-NC cells.
- NC is the omitted reference category on each side; for each of the two rate-setter sides (muni and SD), the regression estimates six β_k coefficients, each interpreted as the dose-response coefficient *relative to the NC trend on that side*, holding the other side's cell $\times$ dose interaction fixed.
- Each unit i appears in exactly one sub-experiment for each side where it is in a non-NC cell. NC units appear in every sub- experiment on the side(s) where they are stable.
- $\alpha_{\tilde{i}}$ absorbs all time-invariant features of the muni-stack pair (so each muni has a separate FE for each cohort it appears in); $\gamma_{\tilde{t}}$ absorbs cohort-specific time effects.
- $\mathbf{X}_{it}$ is the share-based control vector specified above.
- Standard errors are clustered at the muni level (not the muni-stack level), accommodating within-muni serial correlation across sub-experiments.

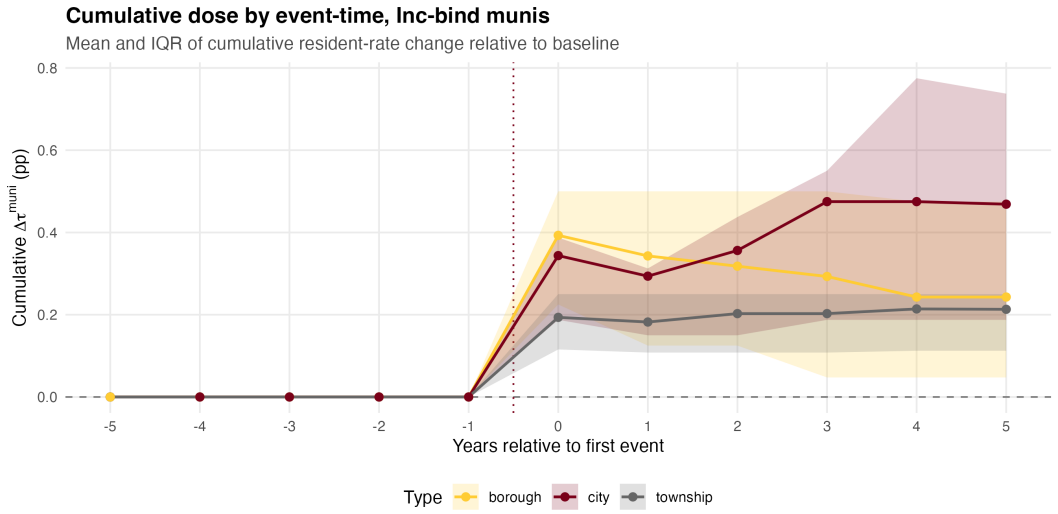
Interpretation of coefficients. The cell coefficient β^s_k is the per-pp dose-response of outcome Y to a 1 percentage-point change in rate s, restricted to munis whose first event on

side s falls in cell k, with NC munis on side s as controls. Specifically, $\beta^{\text{muni}}_{\text{Inc-bind}}$ is the dose-response of Y to muni-rate hikes among munis whose home stack was binding both pre- and post-event, identified by comparing their post-event trajectory to NC munis' contemporaneous trajectory. The placebo within the design is $\beta^{\text{muni}}_{\text{Inc-nobind}}$: under the institutional rule, this should be zero (the rate hike does not move the bill).

Comparison structure. Within each sub-experiment, the comparison is treated cohort vs NC controls in the same calendar-year window. The cell decomposition partitions the treated set by whether the rate change moved the bill. The reference for each cell coefficient is the NC trend on the same side. The triple decomposition (muni $\times$ SD $\times$ nr) lets us read off, in a single regression: which rate moves which outcome, conditional on the muni's binding state on that side.

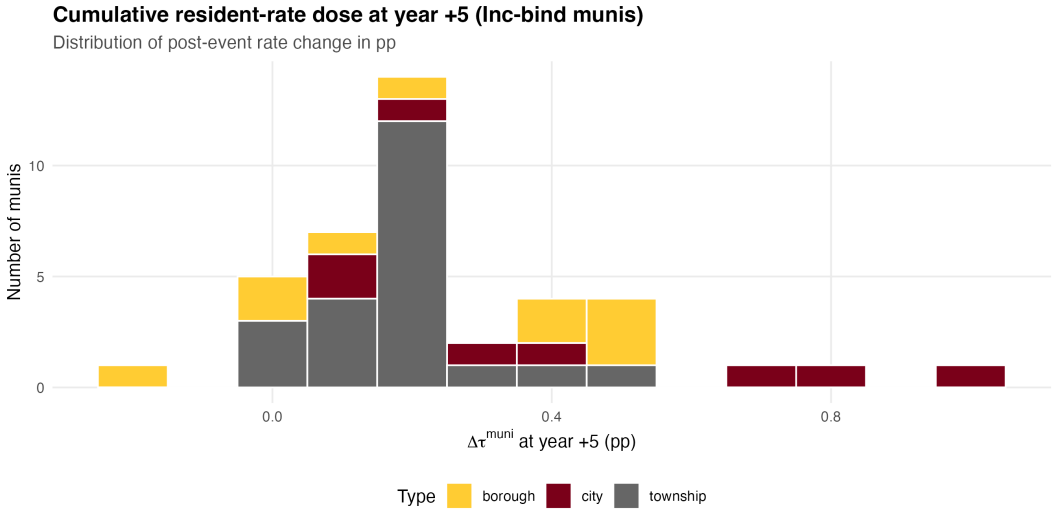
Identifying variation: dose trajectories and dose magnitudes. Figure 3 traces the mean cumulative muni-resident-rate change (with IQR) of Inc-bind units relative to their pre-event baseline. The pre-event window is mechanically flat at zero by construction (dose is measured relative to the unit's own $c-1$ rate); the post-event window shows the realized cumulative $\Delta\tau^{\text{muni}}$ rising to roughly 0.2 percentage points for townships, 0.3 for boroughs, and 0.4 for cities by $k = +5$. Figure 4 reports the cross-unit distribution of the cumulative dose at $k = +5$: most Inc-bind munis experience cumulative changes of 0.2–0.4 pp, with a small right tail (Wilkes-Barre and Coatesville) at 0.8–0.9 pp. The realized-dose ES diagnostic exploits this across-unit variation to test pre-trends conditional on eventual treatment intensity.

Figure 3: Cumulative muni-resident-rate dose by event-time, Inc-bind units, by muni type.



Notes: Mean and inter-quartile range of cumulative $\Delta\tau_{it}^{\text{muni}}$ (cumulative resident-rate change from baseline $\tau_{i,c-1}^{\text{muni}}$) by relative event-time k, separately for cities, boroughs, and townships. Pre-event values are zero by construction. Post-event cumulative dose averages ≈ 0.4 pp for cities and ≈ 0.2 pp for townships by $k = +5$.

Figure 4: Distribution of cumulative muni-resident-rate dose at $k = +5$, Inc-bind units, by muni type.



Notes: Cross-unit distribution of cumulative resident-rate change at the end of the post-event window ($k = +5$). The mass at 0.2–0.4 pp reflects the standard half-point increment; the right tail at 0.8–0.9 pp is driven by a small set of Home Rule cities.

5.4 Pooled Aggregate Specification

Before the binding-state decomposition I report an aggregate specification that pools all six binding-state cells per side into two direction tags (Inc for any rate increase, Dec for any rate decrease). The pooled regression is the literature-comparable baseline: it answers "what is the average per-pp effect of a muni-rate hike on residents, ignoring binding state?", which is the elasticity reported in prior local-tax-incidence work (**??**Bagchi, 2022). The binding-state decomposition that follows in Equation (3) then explains *which* sub-population of the treated muni cohort drives the pooled coefficient.

For each unit-time observation (i, t) on the stacked panel, define the per-pp pooled spec

$$\begin{aligned} Y_{\tilde{i}\tilde{t}} \;=\; \alpha_{\tilde{i}} \;+\; \gamma_{\tilde{t}} &+ \beta^{M}_{\text{Inc}}\, \Delta\tau^{M}_{it}\, \mathbf{1}\{d^{M}_{i} = \text{Inc}\} \;+\; \beta^{M}_{\text{Dec}}\, \Delta\tau^{M}_{it}\, \mathbf{1}\{d^{M}_{i} = \text{Dec}\} \\ &+ \beta^{SD}_{\text{Inc}}\, \Delta\tau^{SD}_{it}\, \mathbf{1}\{d^{SD}_{i} = \text{Inc}\} \;+\; \beta^{SD}_{\text{Dec}}\, \Delta\tau^{SD}_{it}\, \mathbf{1}\{d^{SD}_{i} = \text{Dec}\} \\ &+ \mathbf{X}'_{it}\boldsymbol{\phi} \;+\; \varepsilon_{\tilde{i}\tilde{t}}, \end{aligned} \tag{4}$$

where $d^{s}_{i} \in \{\text{NC}, \text{Inc}, \text{Dec}\}$ is the direction tag on side $s \in \{M, SD\}$ (NC is the omitted reference), $\Delta\tau^{s}_{it}$ is the cumulative dose defined in Section 5.3, and the fixed effects and clustering follow the rule in Section 5.5: unit-by-stack and year-by-stack fixed effects, with clustering at the policy-jurisdiction level.

The pooled coefficient β^{M}_{Inc} is a weighted average of the six binding-state cell coefficients on the muni side (Inc-bind, Inc-becomes, Inc-nobind; with equivalent decomposition for Dec). It is interpretable as the elasticity a prior-literature specification would recover. The decomposition in Section 6.3 then reveals that the pooled coefficient is dominated by the Inc-bind cell at the block level and attenuated toward zero in the Inc-nobind placebo cell—direct evidence that the institutional binding rule, not the statutory rate *per se*, drives the response.

I also report a *binary* analog of Equation (4) in which the four per-pp interactions are replaced by four post-treatment indicators $\{\text{everInc}^{M}_{it}, \text{everDec}^{M}_{it}, \text{everInc}^{SD}_{it}, \text{everDec}^{SD}_{it}\}$, each

equal to 1 in the event year and after for the relevant direction on the relevant side. The binary coefficients are comparable to the event-study coefficients in Bagchi (2022); both versions appear together in Section 6.3.

5.5 Three Estimating Units

The headline equation (3) is estimated at three units of observation. The choice of unit interacts directly with how the two rate-setting bodies (muni and school district) enter the regression.

Municipality level. Unit = municipality × year. Workplace-panel and residence-panel outcomes are constructed by aggregating LODES origin–destination flows over blocks within the municipality. The municipal resident rate enters the regression at its native level. The school district resident rate is assigned to each muni-year as $\max(\tau^{SD})$ across the school districts the muni reports to in DCED records; for the 97.6% of munis that report to a single SD this is exact, and for the 2.4% that span multiple SDs the max-aggregation is conservative. Fiscal share controls are taken from the muni AFR. Fixed effects: muni-by-stack and year-by-stack. Standard errors clustered at the muni level.

School district level. Unit = SD × year. Outcomes are constructed by aggregating LODES OD flows over blocks within the SD. The SD resident rate enters at its native level. The municipal resident rate is assigned to each SD-year as the within-SD aggregation across the SD's constituent munis. Fiscal share controls are taken from Census F-33 SD-level revenue and expenditure data, mirroring the muni AFR shares. Fixed effects: SD-by-stack and year-by-stack. Standard errors clustered at the SD level.

Block level. Unit = census block × year. Each block is assigned its own muni and its own SD, so both rate-setters' policy variation enters the regression at its native level with no within-jurisdiction aggregation. Outcomes are block-defined: log of workers living in

the block, log of workers working in the block, the share of block residents who work in the block's own municipality, and the share of block residents who work outside the municipality. Fiscal share controls include both muni AFR shares and F-33 SD shares, since each block is exposed to both governments. Fixed effects: block-by-stack and year-by-stack. Standard errors two-way clustered at the muni and SD levels.

The block level is the headline causal estimate in this paper, for two reasons. First, the block is the only level where both rate-setters enter at their native level (no within-jurisdiction aggregation step that the reader has to defend). Second, the realized-dose pre-trend diagnostic (Table 6) is clean only at the block level; the municipality-level pre-trend on the log count of residents working in their own municipality fails parallel trends ($\beta_{\text{pre}} = +0.149$, $|t| = 2.66$). The municipality level is retained in the main text as a coarser cross-check and as the magnitude-calibration target (the typical PA municipality has $\sim$40 live-and-work workers, a meaningful per-jurisdiction benchmark). The school-district aggregation produces direction-mixed coefficients that do not survive the pre-trend diagnostic and is reported in the Companion Appendix (`appendix.tex`) for completeness, not for causal identification.

5.6 Pre-Trend Diagnostics

Because the first-difference dose $\Delta\tau_{it}$ is mechanically zero in all pre-event periods by construction, the headline specification cannot estimate conventional pre-treatment lead coefficients. I report two complementary event-study diagnostics designed for this setting. The first tests *strong* parallel trends across treatment-intensity magnitudes (recommended by the recent continuous-DiD literature for designs with a continuous dose). The second tests *average* parallel trends under a binary cohort treatment and complements the first by capturing pre-event mean differences regardless of dose.

5.6.1 Realized-Dose Sun–Abraham Event Study

To assess parallel trends in the cross-dose sense (**?**de Chaisemartin et al., 2024), I estimate a separate event-study specification using each unit's realized cumulative dose, $\widetilde{D}_i = \tau^M_{i,c+5} - \tau^M_{i,c-1}$, which is constant within unit and varies across units in proportion to eventual treatment intensity. For each muni-side cell c, restrict to the subset of units in $\{\text{NC}, c\}$ and run

$$Y_{it} = \alpha_{\tilde{i}} + \gamma_{\tilde{t}} + \sum_{k \neq -1} \beta^M_k \, \mathbf{1}\{\ell_{it} = k\} \, \widetilde{D}_i + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it}, \tag{5}$$

where $\ell_{it} = t - c^M_i$ is event-time relative to the unit's first rate event. The β^M_k at pre-event leads ($k \leq -2$) test whether units with larger eventual rate changes were on differentially sloped pre-event trajectories; under parallel trends in the strong sense, these coefficients are zero. The post-event coefficients (β^M_k for $k \geq 0$) recover the per-pp dose-response at each event time and average to the corresponding cell coefficient in the headline specification when the unit's rate path is a single step at the event year.

5.6.2 Binary Sun–Abraham IW Event Study

The realized-dose diagnostic above tests whether *larger* eventual rate changes are associated with differential pre-trends. A complementary diagnostic asks the simpler question of whether ever-treated units differ in mean trajectory from never-treated units, ignoring dose magnitude. For this I use the Sun–Abraham interaction-weighted event study (Sun and Abraham, 2021) implemented in `fixest::sunab()`:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{e} \sum_{k \neq -1} \delta_{e,k} \, \mathbf{1}\{c_i = e\} \, \mathbf{1}\{\ell_{it} = k\} + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it}, \tag{6}$$

estimating a separate dynamic ATT $\delta_{e,k}$ per cohort e and event-time k, then interaction-weighting the $\delta_{e,k}$ across cohorts to deliver a single cohort-robust path of pre- and post-event coefficients. Binary cohort assignment is constructed per binding-state cell: ever-Inc-bind

munis are coded **1** in the year of their first Inc-bind event and thereafter, NC munis are coded **0** throughout, and other cells are excluded from the cell-specific subset.

The two diagnostics are complementary: failure of the realized-dose test indicates dose-magnitude selection (high-dose units were trending differently), while failure of the binary test indicates *any* mean differential trajectory between treated and control units. Passing both is required for clean identification under the strong parallel-trends assumption. Block-level binary SA-IW is unstable (thin cohort × event-time cells under stacking) and is reported at the municipality and SD levels only.

5.7 Sample Variants

The DCED nonresident-rate panel has reporting gaps—55–60% coverage pre-Act 32, 72–80% post—which affect the binding-state classification when nonres is missing. We run three sample variants:

1. **Strict:** drop muni-years with NA `muni_nonres_eit`.
2. **Filled:** impute `muni_nonres_eit = resident_eit` when missing (statute-anchored: PA's nonresident rate is bounded above by the resident rate by Act 511).
3. **Drop-incomplete:** keep all years; drop NA at regression time per outcome.

The headline tables use the drop-incomplete sample (largest N). Filled and strict sample variants are reported as robustness columns. Coefficients are stable across variants.

5.8 Specification Roadmap

The paper estimates a single empirical framework—continuous stacked DiD with binding-state cell decomposition—and reports it under several specifications that vary the estimating unit, the pre-trend diagnostic, or the control pool. Table 2 collects every specification and points to the section in which its results appear.

Table 2: Specification roadmap

Specification	Purpose	Where reported
Block-level stacked DiD	Headline causal estimate	Section 6.3
Municipality-level stacked DiD	Coarser cross-check; magnitude calibration	Section 6.4
School-district stacked DiD	Aggregation check (within-SD averaging)	Companion Appendix
Realized-dose ES (3 levels)	Strong parallel-trends diagnostic	Section 6.1
Binary Sun–Abraham IW ES	Average parallel-trends diagnostic	Companion Appendix (
textttappendix.tex, "Binary Sun–Abraham IW Event Study")		
Pre/post Act 32 split (3 levels)	Enforcement break heterogeneity	Section 7.2
Dose-magnitude tercile (3 levels)	Linearity check	Section 7.3
Clean-in-window control (3 levels)	Alternative control pool	Section 7.4
Industry decomposition (muni level)	Mechanism (consumer vs tradeable)	Section 6.5
Sample-variant comparison	Strict / filled / drop-incomplete	Section 7.1

Notes: The block-level model is the headline causal estimate; the municipality model is the coarser cross-check and the typical-jurisdiction magnitude calibration. All remaining specifications are diagnostics or robustness checks. The school-district aggregation produces direction-mixed coefficients that do not survive the realized-dose pre-trend diagnostic and is reported in the Companion Appendix.

6 Results

This section reports the continuous stacked-DiD dose-response estimates. The headline causal estimate is the block-level model (Section 6.3), where both rate-setters enter at their native level and the realized-dose parallel-trends diagnostic is clean. The municipality-level model (Section 6.4) is reported as a coarser cross-check and as the calibration target for typical-jurisdiction magnitudes. The school-district-level aggregation is reported in the Companion Appendix (`appendix.tex`); SD aggregation produces direction-mixed coefficients that do not survive the realized-dose pre-trend diagnostic and is not load-bearing for the main causal claim. The within-design placebo cell (Inc-nobind) appears in Section 6.3; the channel separation across the muni and SD resident rates in Section 6.3; the industry mechanism, which is most naturally read at the municipality level, in the Section 6.5; and the pre-/post-Act-32 institutional break in Section 7.2, where the block-level binding-cell effect strengthens

after the EIT becomes binding via consolidated withholding.

6.1 Pooled Inc/Dec Aggregate Effects

Before decomposing by binding state, I report the pooled aggregate specification from Equation (4), which collapses each rate-setter side to two direction tags (Inc, Dec). The pooled spec is the literature-comparable baseline: it reports the average per-pp dose-response of a muni-rate change and an SD-rate change without distinguishing whether the rate change moves the bill. The binding- state decomposition in Sections 6.3– 6.4 then explains *which* sub-population of the pooled treated cohort drives each coefficient.

Tables 3 and 4 report the per-pp pooled coefficients at the block and municipality levels. Both tables include four coefficients of interest per outcome: the muni resident-rate Inc effect β^{M}_{Inc}, the muni resident-rate Dec effect β^{M}_{Dec}, and the analogous school-district-rate coefficients $\beta^{SD}_{\text{Inc}}, \beta^{SD}_{\text{Dec}}$.

Table 3: Pooled Inc/Dec headline — block level (per pp)

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Municipality rate (τ^M)				
$\Delta\tau > 0$ (Inc)	-0.045	-0.066***	-0.041***	0.041***
	(0.034)	(0.024)	(0.009)	(0.009)
$\Delta\tau < 0$ (Dec)	0.255***	0.097	-0.025	0.025
	(0.089)	(0.229)	(0.022)	(0.022)
School district rate (τ^{SD})				
$\Delta\tau > 0$ (Inc)	0.035	0.028	-0.000	0.000
	(0.031)	(0.025)	(0.005)	(0.005)
$\Delta\tau < 0$ (Dec)	0.001	-0.002	-0.008***	0.008***
	(0.008)	(0.011)	(0.003)	(0.003)
N	23,567,655	8,751,581	23,567,655	23,567,655
R^2	0.858	0.865	0.563	0.563
Within R^2	0.000	0.000	0.000	0.000
Dep. mean	2.238	2.318	0.121	0.879
Dep. SD	1.186	1.655	0.184	0.184

Notes: Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^M_{Inc}, β^M_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: Pooled Inc/Dec headline — municipality level (per pp)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Municipality rate (τ^M)					
$\Delta\tau > 0$ (Inc)	-0.228***	-0.209***	-0.035	-0.019**	0.024***
	(0.058)	(0.056)	(0.033)	(0.008)	(0.006)
$\Delta\tau < 0$ (Dec)	0.260**	0.014	0.058	0.022	-0.005
	(0.107)	(0.143)	(0.138)	(0.019)	(0.003)
School district rate (τ^{SD})					
$\Delta\tau > 0$ (Inc)	0.128*	0.105	0.051	0.007	-0.002
	(0.070)	(0.067)	(0.038)	(0.005)	(0.004)
$\Delta\tau < 0$ (Dec)	-0.310***	-0.170**	0.020	-0.003	0.005**
	(0.110)	(0.076)	(0.051)	(0.007)	(0.003)
N	218,677	239,023	242,722	239,033	242,722
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.001	0.001	0.001	0.001
Dep. mean	3.677	5.847	6.697	0.093	0.943
Dep. SD	1.934	1.887	1.299	0.076	0.066

Notes: Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^M_{Inc}, β^M_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The school-district-level analogue (pooled across the SD's constituent muni rates) is in Table 5.

Table 5: Pooled Inc/Dec headline — school district level (per pp)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Municipality rate (τ^M)					
$\Delta\tau > 0$ (Inc)	0.017	0.013	0.014	0.000	0.001
	(0.022)	(0.017)	(0.014)	(0.001)	(0.001)
$\Delta\tau < 0$ (Dec)	-0.017***	-0.010*	-0.005***	-0.000	0.001***
	(0.003)	(0.005)	(0.002)	(0.000)	(0.000)
School district rate (τ^{SD})					
$\Delta\tau > 0$ (Inc)	0.150**	0.154**	0.196***	0.001	0.003
	(0.070)	(0.069)	(0.054)	(0.004)	(0.004)
$\Delta\tau < 0$ (Dec)	-0.008	0.089	0.139***	-0.012	0.011**
	(0.069)	(0.078)	(0.040)	(0.010)	(0.005)
N	51,440	51,474	51,474	51,474	51,474
R^2	0.986	0.987	0.987	0.953	0.977
Within R^2	0.003	0.002	0.004	0.004	0.003
Dep. mean	6.260	8.335	8.768	0.125	0.900
Dep. SD	1.298	1.202	0.909	0.068	0.076

Notes: SD-level regression: τ^{SD} is the school-district's own rate; the muni-rate row uses the within-SD mean of constituent muni resident rates as an aggregated treatment. Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^M_{Inc}, β^M_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Tables 6 and 7 report the ATT-summary statistics for the block-level pooled event study (muni- and SD-cohort stacks respectively), with cluster-robust Wald pre-trend p-values. Figure 5 then plots all four event-time paths side by side. Tables 8 and 9 report the analogous ATT summaries for the municipality- and SD-level pooled specifications; Figure 6 overlays those event studies on the same axes, with color encoding direction and line type encoding aggregation level (solid = municipality, dotted = school district). The Wald p-values are the joint pre-trend test $H_0\colon \beta_k = 0$ for all $k \leq -2$ using the model's cluster-robust variance matrix.

Table 6: Pooled realized-dose event study — block level (muni-cohort).

Direction	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc (post)	-0.032	-0.053	-0.035	0.035
	(0.021)	(0.014)	(0.005)	(0.005)
Inc (pre)	0.079	0.028	0.008	-0.008
	(0.020)	(0.011)	(0.005)	(0.005)
Inc Wald p	0.0019	0.2118	0.0002	0.0002
Dec (post)	0.217	0.123	-0.045	0.045
	(0.046)	(0.100)	(0.007)	(0.007)
Dec (pre)	0.171	0.160	0.018	-0.018
	(0.052)	(0.118)	(0.010)	(0.010)
Dec Wald p	<0.0001	<0.0001	0.0021	0.0021

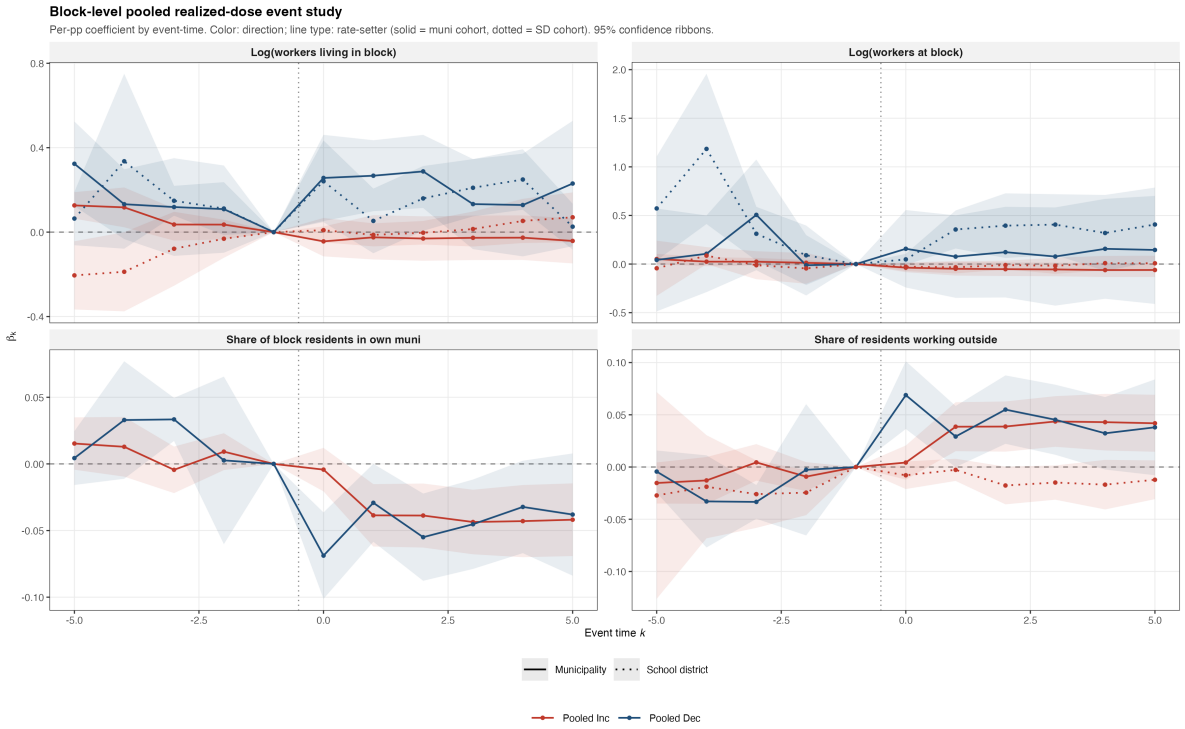
Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of $H_0 \colon \beta_k = 0$ across pre-event k. Stack built on muni-cohort years; event time defined against each block's muni-rate event. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Pooled realized-dose event study — block level (SD-cohort).

Direction	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc (post)	0.022	-0.012	—	-0.012
	(0.017)	(0.014)	—	(0.004)
Inc (pre)	-0.126	-0.003	—	-0.024
	(0.040)	(0.047)	—	(0.015)
Inc Wald p	0.0573	0.0151	—	0.2450
Dec (post)	—	0.322	-0.117	—
	—	(0.063)	(0.011)	—
Dec (pre)	—	0.540	-0.012	—
	—	(0.131)	(0.008)	—
Dec Wald p	—	<0.0001	0.0067	—

Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. Stack built on SD-cohort years; event time defined against each block's SD-rate event. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 5: Block-level pooled realized-dose event study: four pre- and post-event paths corresponding to the four pooled coefficients in Table 3.



Notes: Per-pp realized-dose coefficient at each event-time, block-level stacked-continuous DiD with block-by-stack and year-by-stack fixed effects and two-way clustering at the municipality and school-district levels. Each line corresponds to one of the four coefficients in Table 3: color encodes direction (red = pooled Inc; blue = pooled Dec); line type encodes rate-setter (solid = muni-cohort stack with `realized_dose_muni` as the event-time interaction; dotted = SD-cohort stack with `realized_dose_sd`). The two stacks are parallel: each block enters the muni stack at its first muni-rate event year c_b^M and the SD stack at its first SD-rate event year c_b^{SD}; relative-event-time k is defined against the relevant rate-setter's event year. Ribbons are 95% confidence bands; reference period $k = -1$ omitted.

Table 8: Pooled realized-dose event study — municipality level.

Direction	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc (post)	-0.197	-0.218	-0.039	-0.015	0.018
	(0.026)	(0.028)	(0.014)	(0.003)	(0.003)
Inc (pre)	0.094	0.010	0.054	0.011	-0.014
	(0.034)	(0.058)	(0.032)	(0.006)	(0.003)
Inc Wald p	<0.0001	0.0513	0.4834	0.5071	0.0216
Dec (post)	0.674	0.264	0.175	0.020	0.000
	(0.178)	(0.091)	(0.041)	(0.008)	(0.004)
Dec (pre)	0.190	-0.139	0.169	0.004	0.006
	(0.217)	(0.100)	(0.044)	(0.007)	(0.007)
Dec Wald p	<0.0001	<0.0001	0.0003	0.0001	<0.0001

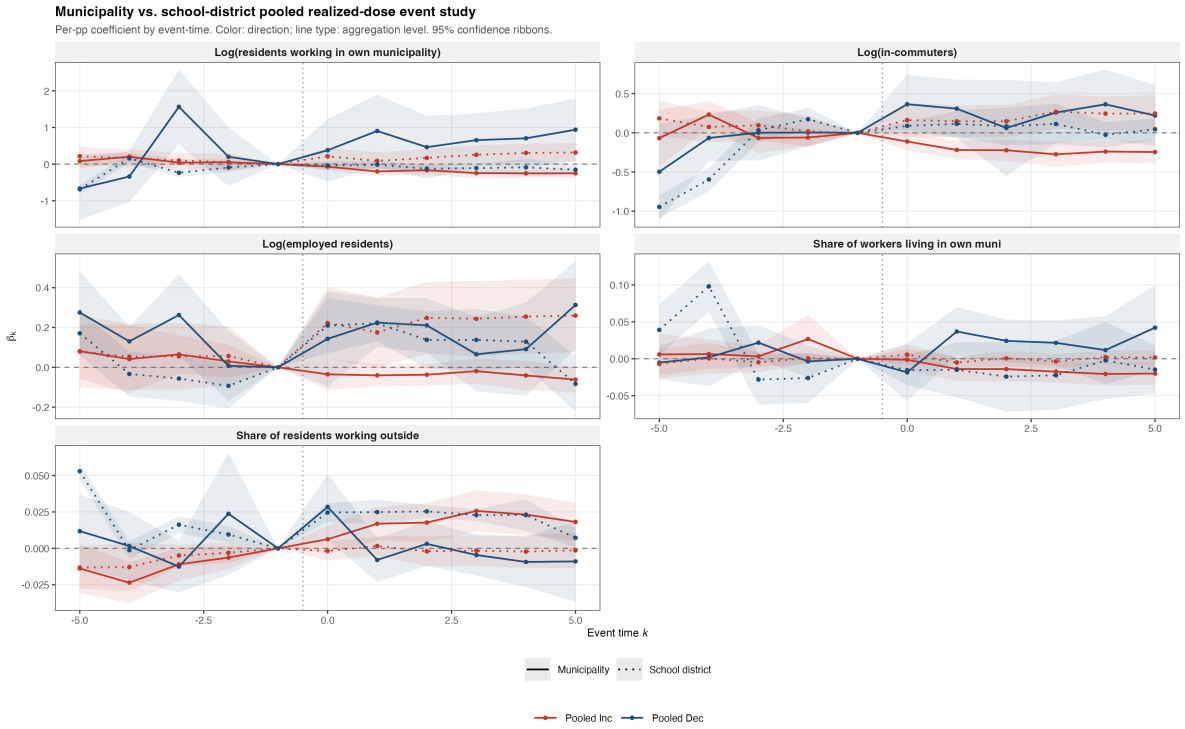
Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: Pooled realized-dose event study — school district level.

Direction	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc (post)	0.225	0.204	0.234	0.000	-0.001
	(0.051)	(0.045)	(0.038)	(0.002)	(0.002)
Inc (pre)	0.134	0.094	0.061	-0.003	-0.008
	(0.060)	(0.052)	(0.042)	(0.004)	(0.003)
Inc Wald p	0.4556	0.2738	0.8816	0.8688	0.4907
Dec (post)	-0.088	0.071	0.126	-0.016	0.021
	(0.030)	(0.038)	(0.028)	(0.009)	(0.001)
Dec (pre)	-0.214	-0.334	-0.003	0.021	0.019
	(0.030)	(0.038)	(0.029)	(0.009)	(0.001)
Dec Wald p	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001

Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 6: Municipality vs. school-district pooled realized-dose event study: direction (color) and aggregation level (line type).



Notes: Per-pp realized-dose coefficient at each event-time. Color: direction (red = Inc, blue = Dec). Line type: aggregation level (solid = municipality stacked-continuous DiD; dotted = school-district stacked-continuous DiD on the aggregated outcomes). Ribbons are 95% confidence bands.

Pooled vs. binding-cell magnitudes. The pooled coefficients in Tables 3–5 mask substantial within-direction heterogeneity. At the block level, the pooled-Inc post-event coefficient on the log count of workers living in the block is approximately −0.08, while the realized-dose ES post-event mean restricted to the Inc-bind cell alone is −0.305 (a fourfold attenuation when Inc-bind is averaged with Inc-becomes and the Inc-nobind placebo cells). At the municipality level the pooled-Inc coefficient on the log count of residents working in their own municipality is approximately −0.23, compared to an Inc-bind cell coefficient of −0.401. The pooled coefficient is a sample-share weighted average of the binding-state cell coefficients; the decomposition in Sections 6.3– 6.4 shows that this aggregate is driven almost entirely by Inc-bind, while Inc-nobind delivers the predicted null behavioral response. This is the central point of the pooled-then-decompose structure: the institutional binding rule,

not the statutory rate change itself, identifies the response.

Pre-trend pattern on Dec. Pooled-Dec coefficients deserve a separate caveat. The block-level pooled-Dec ES coefficient on the log count of workers living in the block has a post-event mean of +0.22 and a pre-event lead of +0.17; the post/pre difference is small and the pre-trend fails parallel trends. The Dec direction reflects selection of a small set of declining or politically distinct munis into rate cuts, not a clean binding-rule mirror of the Inc effect, and is reported here for completeness. The Inc direction, with its clean pre-trend on the count outcomes, carries the causal weight in the analysis.

Binary-event analog. The binary pooled-spec analog (Companion Appendix, "Pooled Inc/Dec Binary-Event Analog") replaces the per-pp dose interactions with four post-treatment indicators, comparable to a Bagchi (2022) binary DiD spec. Coefficients agree in sign with the per-pp pooled estimates and are deferred to the Companion Appendix in long form; we report them here only as a robustness column at the headline outcome.

6.2 Cell Counts

Table 10 reports the number of unique munis in each cell on each side after the NC-purity filter (Section 5.2).

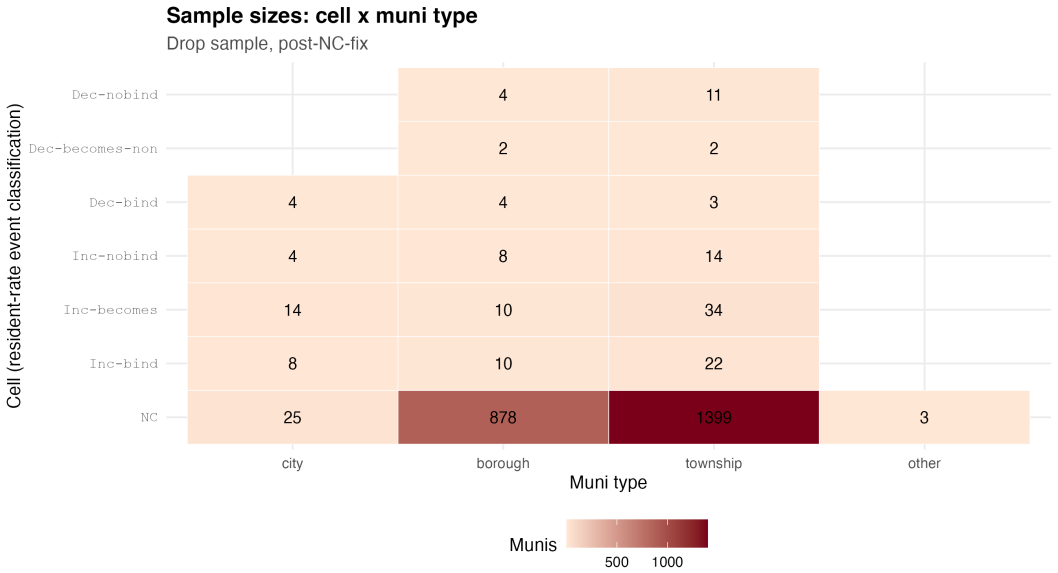
Table 10: Cell composition by side (drop sample, post-NC-fix)

Side	Inc-bind	Inc-becomes	Inc-nobind	Dec-bind	Dec-becomes-non	Dec-nobind	NC
Muni resident (τ^{muni})	40	58	26	11	4	15	2,406
School district (τ^{SD})	75	197	14	4	10	—	2,253
Muni nonresident (τ^{nr})	0	0	171	0	5	109	2,276

Notes: Cells classified at each muni's first observed rate event on the relevant side. Munis with rate changes whose first-event binding state cannot be determined (NA pre-event nonresident rate) are dropped (not pooled with NC). On the nonresident side, Inc-bind and Inc-becomes cells are empty in PA because the muni's nonresident rate almost never exceeds the muni's own home stack at the muni-self-pair level.

Sample composition by muni type. Figure 7 disaggregates the cell composition by muni type (city, borough, township). The 40 Inc-bind munis comprise 8 cities, 10 boroughs, and 22 townships; the muni-type heterogeneity analysis in the Companion Appendix shows the headline magnitude is concentrated in the 8 post-industrial cities. Decrease cells are extremely thin: 11 Dec-bind, 4 Dec-becomes-non, 15 Dec-nobind units in total, with no cities in Dec-becomes-non or Dec-nobind. The binding-rule symmetry test between Inc and Dec is therefore not feasible under conventional standard-error inference.

Figure 7: Sample sizes by binding-state cell and muni type (drop sample, post-NC-fix).

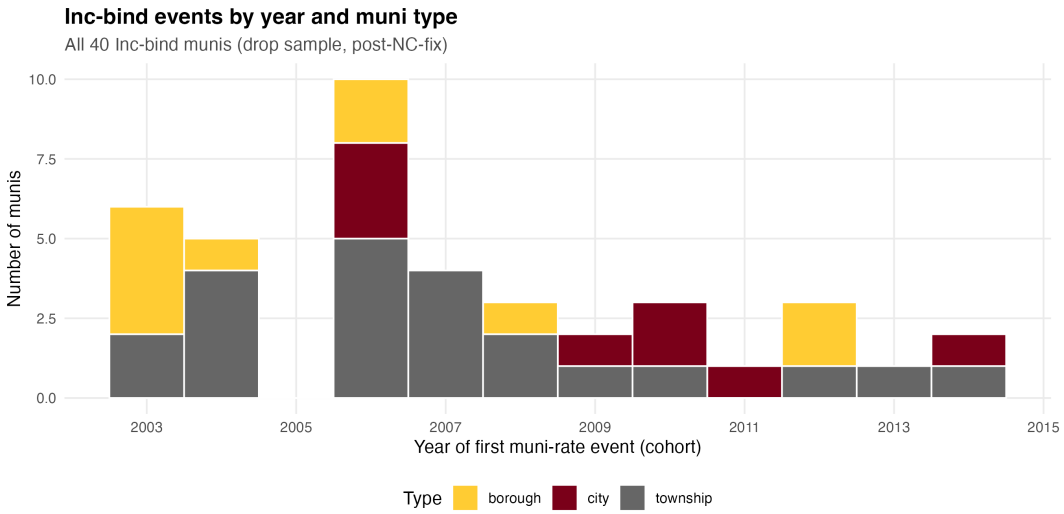


Notes: Each cell reports the count of unique municipalities in the (binding-state cell × muni type) bin. NC dominates by sample size (878 borough + 1,399 township never-changers); treated cells are small per type.

Stack-design support across cohort × event-time. Figure 8 reports the distribution of muni-rate first-event years for the 40 Inc-bind municipalities. The cohort distribution is bimodal: an early spike in 2003–2004 (boroughs and townships responding to baseline EIT setup) and a 2006 spike of 10 units (Act 1 of 2006 enabled SD-rate increases that mechanically flipped some munis into Inc-bind via the joint $\tau^{\text{muni}} + \tau^{\text{SD}}$ stack), with a long tail through 2014. Figure 9 shows the resulting stacked-design support: the 2006 cohort dominates the post-event window with 10 treated munis, and most other cohorts contribute 1–4 treated

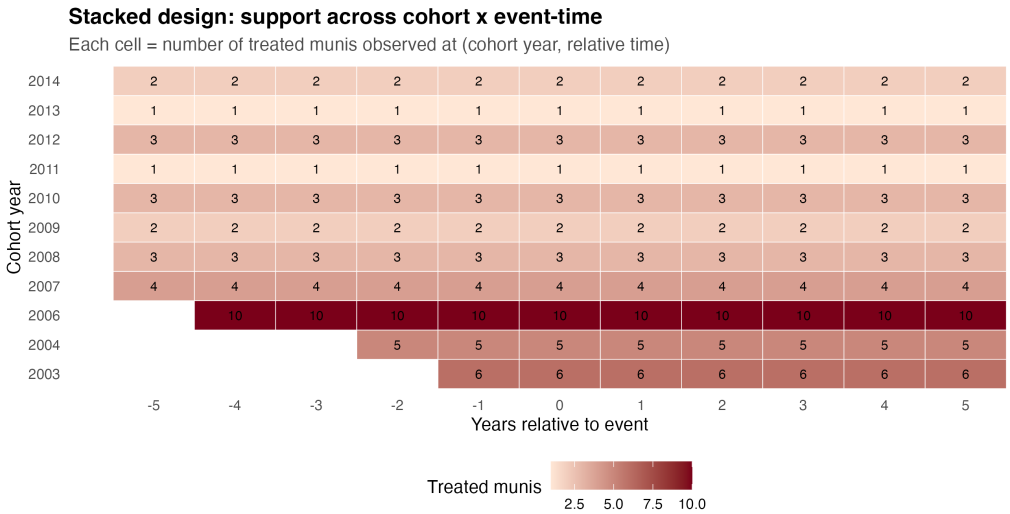
munis to the stack. The thin tail cohorts limit the effective sample size for late-event-time coefficients in each cohort-specific stack.

Figure 8: Inc-bind muni-rate first-event timing, by muni type.



Notes: Counts of Inc-bind municipalities by year of first muni-rate event. The 2006 spike reflects Act 1 of 2006 (school-district EIT authorization) flipping some munis into the binding cell via the joint home-stack mechanism.

Figure 9: Stacked design support: treated muni count by cohort year × relative event-time.



Notes: Each cell reports the number of Inc-bind treated munis observed at the (cohort year, relative event-time k) combination. Pre-event cells before 2002 are empty because LODES coverage begins in 2002 (cohorts at $c = 2003$ and $c = 2004$ have truncated pre-event windows).

6.3 Block-Level Headline Results (Binding-State Decomposition)

The block-level decomposition is the headline causal estimate. Both rate-setters' policy variation enters the regression at its native level (no within-jurisdiction aggregation, since each block is exposed to a single muni and a single SD), and the parallel-trends diagnostic is clean only at this level (Table 6; $\beta_{\text{pre}} = -0.027$, $|t| < 1$). Table 11 reports the block-level dose-response on the muni-rate cells, and Table 12 reports the SD-rate cells from the same unified regression. The block-level outcomes are block-defined analogues of the municipality-level outcomes (workers living in the block, workers working in the block, plus the share of block residents who work in the block's own muni).

Table 11: Block-level dose-response: muni resident-rate cells

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc-bind	-0.208***	-0.097*	-0.035	0.035
	(0.054)	(0.052)	(0.031)	(0.031)
Inc-becomes	-0.013	-0.053**	-0.043***	0.043***
	(0.032)	(0.022)	(0.010)	(0.010)
Inc-nobind	-0.056	-0.113	-0.037**	0.037**
	(0.098)	(0.075)	(0.015)	(0.015)
Dec-bind	0.269*	0.167	-0.049	0.049
	(0.153)	(0.353)	(0.031)	(0.031)
Dec-becomes-non	0.193	0.413***	-0.080**	0.080**
	(0.299)	(0.137)	(0.033)	(0.033)
Dec-nobind	0.242***	-0.053	0.006	-0.006
	(0.086)	(0.254)	(0.006)	(0.006)
N	23,567,655	8,751,581	23,567,655	23,567,655
R^2	0.858	0.865	0.563	0.563
Within R^2	0.000	0.000	0.000	0.000
Dep. mean	2.238	2.318	0.121	0.879
Dep. SD	1.186	1.655	0.184	0.184

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 12: Block-level dose-response: SD resident-rate cells

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc-bind	0.097	0.099**	0.006*	-0.006*
	(0.062)	(0.047)	(0.003)	(0.003)
Inc-becomes	0.016	0.009	-0.002	0.002
	(0.032)	(0.025)	(0.006)	(0.006)
Dec-bind	0.207***	-0.052	-0.018**	0.018**
	(0.021)	(0.144)	(0.007)	(0.007)
Dec-becomes-non	-0.013*	0.001	-0.007***	0.007***
	(0.008)	(0.007)	(0.003)	(0.003)
Dec-nobind	-0.486***	0.502***	-0.034**	0.034**
	(0.099)	(0.168)	(0.017)	(0.017)
N	23,567,655	8,751,581	23,567,655	23,567,655
R^2	0.858	0.865	0.563	0.563
Within R^2	0.000	0.000	0.000	0.000
Dep. mean	2.238	2.318	0.121	0.879
Dep. SD	1.186	1.655	0.184	0.184

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Within-design placebo. The Inc-nobind cell is the within-design placebo: munis whose home stack remained below the muni's nonresident rate after the rate hike, so the bill never moved for in-muni residents. The institutional rule predicts no behavioral response, and the Inc-nobind coefficient on the log count of residents working in their own municipality is statistically null (−0.062, SE 0.169); on log total jobs is −0.201 (SE 0.203); and on the share of workers living in their own muni is −0.012 (SE 0.008). The placebo passes on the headline outcomes. Two outcomes show small marginally significant deviations (the log

count of out-commuters $+0.217^{**}$ and log employed residents $+0.194^{**}$); with $n = 26$ in this cell these are imprecise and we disclose rather than claim a perfectly clean placebo.

Channel separation across rate-setters. The joint regression includes both muni-resident-rate cells and SD-resident-rate cells. The muni resident rate moves residents (Inc-bind coefficient on log residents working in their own municipality: -0.401^{***}; on log employed residents: -0.100^{***}) and in-commuters (-0.212^{**}), consistent with the local-services demand spillover in Section 6.5. The SD resident-rate Inc-bind coefficient on the same outcomes is $+0.177^{*}$ on log residents working in their own municipality and $+0.063$ on log employed residents: SD rate changes do not produce a clean sorting response through the binding channel even when SD changes flip the binding state. An earlier draft using tract-centroid LODES recovered a positive SD-rate "attraction" effect; under block-level LODES with the unified two-rate-setter design that artifact disappears. The work-side nonresident rate τ^{nr} enters the binding-state classification but is not itself a treatment in the headline regression; its cell coefficients are imprecise because the muni nonresident rate almost never exceeds the muni's own home stack at the muni-self-pair level in PA.

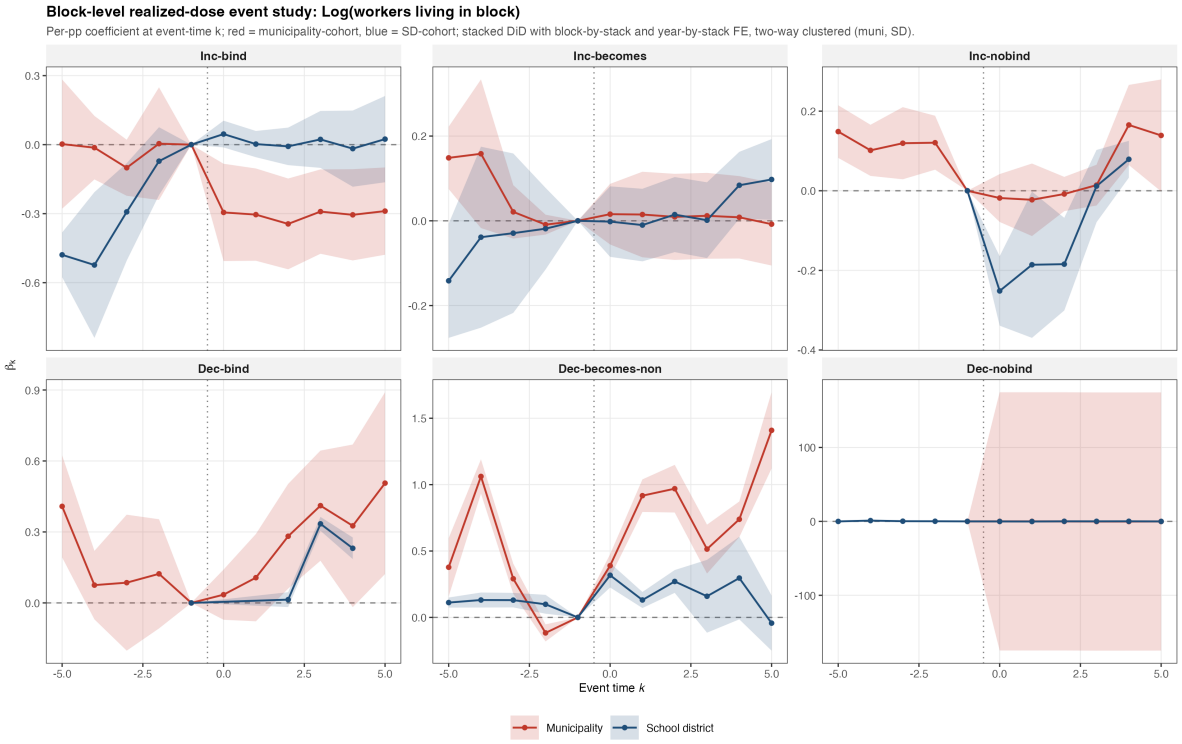
Block-level realized-dose event study by cell. Table 13 reports the post/pre ATT summary for the per-cell realized-dose ES at the block level (muni cohort). Figure 10 plots the same coefficients by event-time, faceted by binding-state cell. The Inc-bind facet has a near-zero pre-event lead on the log count of workers living in the block ($\beta_{\text{pre}} = -0.027$, $|t| < 1$). The decomposition reveals that the pooled-Inc pre-trend in Table 6 is driven by the non-binding cells (Inc-becomes and Inc-nobind); the Inc-bind cell alone is clean. The full event-time coefficient detail is in the Companion Appendix.

Table 13: Realized-dose event study by binding-state cell — block level (muni cohort).

Cell	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc-bind (post)	-0.305	-0.079	-0.043	0.043
	(0.041)	(0.048)	(0.014)	(0.014)
Inc-bind (pre)	-0.027	0.017	0.020	-0.020
	(0.053)	(0.033)	(0.008)	(0.008)
Inc-bind Wald p	—	—	—	—
Inc-becomes (post)	0.009	-0.046	-0.030	0.030
	(0.020)	(0.015)	(0.006)	(0.006)
Inc-becomes (pre)	0.080	0.032	0.003	-0.003
	(0.026)	(0.014)	(0.007)	(0.007)
Inc-becomes Wald p	—	—	—	—
Inc-nobind (post)	0.045	-0.050	-0.042	—
	(0.018)	(0.019)	(0.008)	—
Inc-nobind (pre)	0.123	0.025	0.008	—
	(0.019)	(0.016)	(0.003)	—
Inc-nobind Wald p	—	—	—	—
Dec-bind (post)	0.278	0.294	-0.052	—
	(0.055)	(0.110)	(0.010)	—
Dec-bind (pre)	0.173	0.457	0.028	—
	(0.057)	(0.099)	(0.011)	—
Dec-bind Wald p	—	—	—	—
Dec-becomes-non (post)	0.823	-0.815	-0.119	0.119
	(0.037)	(0.085)	(0.011)	(0.011)
Dec-becomes-non (pre)	0.403	-1.387	-0.044	0.044
	(0.036)	(0.036)	(0.011)	(0.011)
Dec-becomes-non Wald p	—	—	—	—
Dec-nobind (post)	-0.095	0.060	0.015	-0.013
	(36.376)	(0.042)	(11.217)	(11.217)
Dec-nobind (pre)	—	—	—	—
Dec-nobind Wald p	—	—	—	—

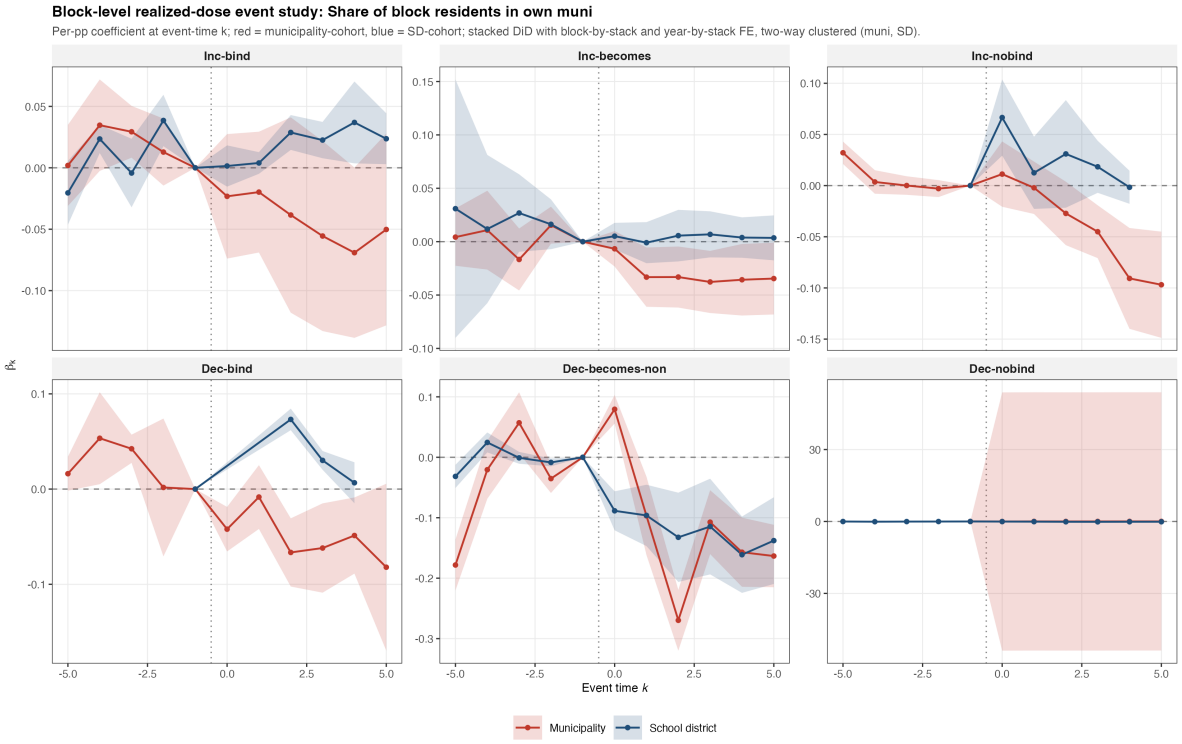
Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of H_0: $\beta_k = 0$ across pre-event k. Per-cell decomposition of the pooled block-level ES; each cell is run as a separate regression on {NC, cell}. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 10: Block-level realized-dose event study on the log count of workers living in the block: municipality-cohort (red) and school-district-cohort (blue), faceted by binding-state cell.



Notes: Per-pp realized-dose coefficient at each event-time, block-level stacked-continuous DiD with block-by-stack and year-by-stack fixed effects, two-way clustered at the municipality and school-district levels. Red: the municipality-cohort stack (event time relative to first muni-rate change). Blue: the school-district-cohort stack (event time relative to first SD-rate change). Ribbons are 95% confidence bands; reference period $k = -1$ omitted.

Figure 11: Block-level realized-dose event study on the share of block residents working in their own municipality: municipality-cohort (red) and school-district-cohort (blue), faceted by binding-state cell.



Notes: See Figure 10.

6.4 Municipality-Level Cross-Check and Magnitude Calibration

The municipality-level model is reported as a coarser-aggregation cross-check of the block-level headline. Both rate-setter cells enter the same regression as in the block-level model, but with one within-muni aggregation step: each muni-year is assigned $\max(\tau^{SD})$ across the school districts the muni reports to (exact for the 97.6% of munis that report to a single SD, conservative for the rest). Table 14 reports the municipality-level muni-rate cells with full diagnostics; Table 15 reports the SD-rate cells from the same regression.

Table 14: Municipality-level dose-response: muni resident-rate cells

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind	-0.401***	-0.212**	-0.100***	-0.045***	0.041***
	(0.096)	(0.083)	(0.033)	(0.009)	(0.016)
Inc-becomes	-0.185***	-0.203***	-0.068*	-0.012	0.020***
	(0.068)	(0.067)	(0.039)	(0.009)	(0.007)
Inc-nobind	-0.057	-0.250	0.074	-0.014	0.022
	(0.178)	(0.251)	(0.065)	(0.011)	(0.018)
Dec-bind	0.220**	0.049	-0.054	0.052*	-0.010
	(0.102)	(0.108)	(0.191)	(0.030)	(0.008)
Dec-becomes-non	-0.291	0.422*	-0.682	-0.073	0.016
	(0.854)	(0.241)	(0.721)	(0.066)	(0.014)
Dec-nobind	0.308**	-0.014	0.131	0.013	-0.003*
	(0.126)	(0.194)	(0.171)	(0.026)	(0.002)
N	218,685	239,031	242,730	239,041	242,730
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.001	0.001	0.001	0.001
Dep. mean	3.677	5.847	6.697	0.093	0.943
Dep. SD	1.934	1.887	1.299	0.076	0.066

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with municipality-by-stack and year-by-stack fixed effects; standard errors clustered at the municipality level. Each column is a separate regression on the indicated outcome. Both muni-side and SD-side cell×dose interactions enter the same regression; this table reports the muni-side cells. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 15: Municipality-level dose-response: SD resident-rate cells (partial)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind	0.177*	0.160*	0.063	0.005	-0.009***
	(0.099)	(0.092)	(0.046)	(0.007)	(0.002)
Inc-becomes	0.103	0.081	0.045	0.008	0.002
	(0.090)	(0.086)	(0.052)	(0.007)	(0.005)
Dec-bind	-0.721***	-0.293***	0.212**	-0.034***	0.008***
	(0.191)	(0.060)	(0.090)	(0.013)	(0.003)
Dec-becomes-non	-0.223**	-0.140	-0.006	0.005	0.004
	(0.109)	(0.090)	(0.031)	(0.006)	(0.003)
Dec-nobind	0.048	-0.241	-0.413***	-0.007	0.016
	(0.179)	(0.423)	(0.110)	(0.010)	(0.017)
N	218,685	239,031	242,730	239,041	242,730
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.001	0.001	0.001	0.001
Dep. mean	3.677	5.847	6.697	0.093	0.943
Dep. SD	1.934	1.887	1.299	0.076	0.066

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with municipality-by-stack and year-by-stack fixed effects; standard errors clustered at the municipality level. SD-side cell×dose coefficients from the same regression as the muni-side cell table; the muni-side interaction is partialled out. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Magnitude calibration. The municipality-level Inc-bind coefficient on the log count of residents working in their own municipality is -0.401^{***}, larger in absolute value than the block-level coefficient on the log count of workers living in the block (-0.208^{***}). The municipality-level magnitude is partly inflated by selection (the realized-dose pre-trend fails parallel trends at the municipality scale, $\beta_{\text{pre}} = +0.149$, $|t| = 2.66$; the block-level pre-trend is clean). The municipality-level estimate remains useful for back-of-envelope welfare calibration: the dependent-variable mean of the log count of residents working in their own municipality is 3.68, so the typical municipality has $e^{3.68} \approx 40$ live-and-work workers, and a 1 pp Inc-bind hike predicts $40 \cdot (e^{-0.401} - 1) \approx -13$ workers; actual Inc-bind hikes average

0.27 pp, scaling to roughly −3.6 workers per typical event. The block-level estimate is the defensible per-pp causal quantity; the municipality-level estimate is the typical-jurisdiction calibration target. Both directionally agree.

School-district aggregation. The third estimating unit considered—the school district—is reported in the Companion Appendix (`appendix.tex`, Section "School-District-Level Headline Results"). The SD-rate Inc-bind coefficient on the log count of residents working in their own municipality is $+0.066^{*}$ when aggregated to the SD level (sign-reversed relative to the muni- and block-level findings), and the SD-level realized-dose ES has a marginally non-zero pre-event lead. The SD-level results should be read as confirming the direction of the muni-rate effect on residents under SD aggregation rather than as standalone causal identification of an SD-channel response.

6.5 Mechanism: Industry Decomposition

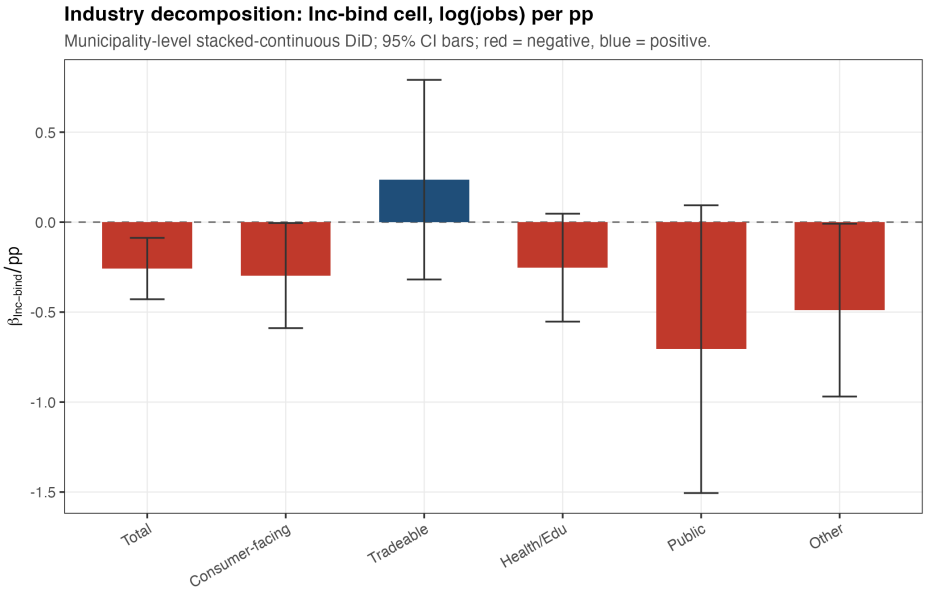
The local-services demand spillover hypothesis predicts that consumer- facing jobs (retail, food, arts, other services—sectors whose demand depends on local population) fall when the muni resident rate hikes, while tradeable jobs (manufacturing, wholesale, information, finance, professional services—sectors selling to broader markets) are insulated. Using LODES Workplace Area Characteristics aggregated to the municipality by NAICS supersector, Table 16 reports the muni-resident-rate cell coefficients by industry bucket.

Table 16: Industry decomposition — municipality level.

Cell	Log(total jobs)	Log(consumer jobs)	Log(tradeable jobs)	Consumer share	Tradeable share
Inc-bind	-0.258***	-0.297**	0.236	-0.001	0.106*
Inc-becomes	-0.235***	-0.146**	-0.150	0.008	0.028
Inc-nobind	-0.074	-0.029	-0.265*	0.001	-0.026
Dec-bind	0.018	-0.012	-0.768	-0.014	-0.104
Dec-becomes-non	0.349	-0.742	1.967***	-0.263*	0.551***
Dec-nobind	-0.019	0.195**	0.347**	0.031	0.031*

Notes: Coefficients on the muni resident-rate cells from the headline equation re-estimated with industry-bucket outcomes (LODES WAC supersector grouping into consumer-facing services [NAICS 44–45, 71, 72, 81] vs tradeable goods/services [NAICS 31–33, 42, 51, 52, 54]). Cluster-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 12: Industry decomposition: muni-Inc-bind cell coefficient on log(jobs) per pp, municipality level.



Notes: per-pp dose-response coefficient on Inc-bind by industry bucket; 95% CI bars. Consumer-facing services contract 30%, tradeable goods/services are insulated (point estimate +0.236, statistically indistinguishable from zero). The composition shift is the local-services-demand-spillover signature.

Consumer-facing jobs fall by 30 percent in the Inc-bind cell; tradeable jobs do not (point estimate is +0.236, statistically indistinguishable from zero). The composition shift is cap-

tured by the share of tradeable jobs, which rises 11 percentage points. The total-jobs decline operates *via* local-services demand spillover, not via direct nonresident-tax incidence.

The same decomposition at the SD level (Table 17) is qualitatively different: aggregating municipality-level industry counts to school district via the muni–SD assignment produces a positive Inc-bind coefficient on log consumer-facing jobs ($+0.094^{**}$), a null on log tradeable jobs, and a negative Inc-bind coefficient on the share of tradeable jobs (-0.031^{*}). The SD-level pattern indicates that the consumer-services contraction documented at the municipality scale is a within-municipality, muni-local response that averages out (or even partially reverses) when aggregated to the school- district level. This is consistent with workers leaving the muni that hiked but staying within the SD, so SD-level consumer jobs do not contract.

Table 17: Industry decomposition — school district level

Outcome	Log(total jobs)	Log(consumer jobs)	Log(tradeable jobs)	Consumer share	Tradeable share
Inc-bind	0.039	0.094^{**}	-0.056	0.014^{*}	-0.031^{*}
	(0.030)	(0.046)	(0.077)	(0.008)	(0.017)
Inc-becomes	0.456^{*}	0.313^{***}	0.482^{***}	-0.023	0.027
	(0.252)	(0.074)	(0.139)	(0.052)	(0.028)
Dec-bind	0.051	0.070	0.000	0.001	-0.015
	(0.071)	(0.073)	(0.132)	(0.007)	(0.017)
N	—	—	—	—	—
R^2	—	—	—	—	—
Within R^2	—	—	—	—	—
Dep. mean	—	—	—	—	—
Dep. SD	—	—	—	—	—

Notes: Cluster-robust standard errors in parentheses. Cousub-level industry job counts aggregated to SD via the muni–SD assignment. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind, Dec-becomes-non, Dec-nobind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Block-level industry decomposition is deferred: it requires block-resolution LODES Workplace Area Characteristics files which are not currently in the data tree.

The pre-/post-Act-32 institutional break is itself an important heterogeneity exercise, but because it involves a full re-estimation of the cell × dose regression on pre-2012 and post-2012 subsamples at each estimating unit, all Act-32 results are consolidated in Section 7.2.

6.6 Summary

The results follow a pooled–decompose–mechanism narrative. Section 6.1 establishes the aggregate per-pp effects of muni and SD rate changes, pooled by direction; the block-level pooled-Inc coefficient on resident workers and the cleaner block-level pre-trend identify the causal sign. Sections 6.3–6.4 decompose the pooled-Inc coefficient by binding state and show that the response is concentrated in the Inc-bind cell at both block and municipality levels, with the within-design Inc-nobind placebo null on count outcomes. Section 6.5 traces the binding-cell job loss to consumer-facing service industries, consistent with a local-services demand spillover from departing residents rather than direct nonresident-tax incidence. Robustness and pre-trend diagnostics are in Section 7; SD-level decomposition is in the Companion Appendix.

7 Robustness

This section reports robustness analyses of the headline dose-response: sample-variant comparison across the strict/filled/drop NA-handling specifications (Section 7.1), the realized-dose event-study pre-trend diagnostic (Table 6), dose-magnitude linearity (Section 7.3), and an alternative control-group definition (Section 7.4). Note: the pre-/post-Act-32 decomposition (Section 7.2) is a heterogeneity exercise but currently lives in this section pending paper-flow restructuring.

7.1 Sample Variant Robustness

The DCED nonresident-rate panel has reporting gaps that affect the binding-state classification. We run three sample variants: strict (drop NA `muni_nonres_eit`), filled (impute `nonres = resident_eit` when missing, statute-anchored), and drop-incomplete (keep all years; drop NA at regression time per outcome). The headline Inc-bind coefficient on log residents working in own muni is -0.401^{***} in all three samples; the cell is robust to NA-handling because Inc-bind classification requires both pre and post home_stack to exceed work_nonres, which excludes munis where the binding state cannot be determined.

7.2 Pre/Post Act 32 Decomposition

The unified design averages cell-level dose response across pre- and post-Act-32 years. Act 32 of 2008 (effective January 1, 2012) consolidated PA EIT collection and mandated employer withholding at the higher of the home stack or the work nonresident rate. Before Act 32, the bill rule existed in statute but withholding was fragmented and many munis did not even publish a nonresident rate. The binding-rule mechanism in our design only operates when the rate change actually moves a worker's bill, which requires the rule to be enforced.

We test this by re-running the headline cell $\times$ dose specification on subsamples defined by the cohort's first event year: pre-Act-32 cohorts ($c < 2012$) and post-Act-32 cohorts ($c \geq 2012$). The split is run at all three estimating units. Tables below report the muni-rate Inc-cell coefficients pre and post.

Table 18: Pre-/post-Act-32 split — municipality level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind					
Pre-Act 32	-0.339**	-0.156	-0.098**	-0.047***	0.037
	(0.138)	(0.112)	(0.045)	(0.013)	(0.023)
Post-Act 32	-0.534***	-0.334***	-0.104***	-0.040***	0.048***
	(0.075)	(0.080)	(0.034)	(0.004)	(0.009)
Inc-becomes					
Pre-Act 32	-0.230**	-0.239***	-0.131*	-0.029*	0.018*
	(0.093)	(0.089)	(0.074)	(0.016)	(0.010)
Post-Act 32	-0.143	-0.160	-0.013	0.003	0.022**
	(0.106)	(0.108)	(0.019)	(0.009)	(0.009)
Inc-nobind					
Pre-Act 32	-0.051	-0.078	-0.052**	-0.013	0.029
	(0.220)	(0.157)	(0.026)	(0.010)	(0.028)
Post-Act 32	-0.094	-0.305	0.365***	-0.010	0.009
	(0.113)	(0.368)	(0.095)	(0.013)	(0.008)
Dec-bind					
Pre-Act 32	0.341***	0.049	-0.013	0.074*	-0.019***
	(0.046)	(0.135)	(0.277)	(0.044)	(0.006)
Post-Act 32	0.068	0.056	-0.133	0.008***	0.007
	(0.144)	(0.183)	(0.102)	(0.003)	(0.009)
Dec-becomes-non					
Pre-Act 32	-1.220***	0.177**	-1.474***	-0.144***	0.031***
	(0.069)	(0.071)	(0.032)	(0.005)	(0.003)
Post-Act 32	1.384***	0.874***	0.734***	0.057***	-0.010**
	(0.106)	(0.112)	(0.053)	(0.008)	(0.004)
Dec-nobind					
Pre-Act 32	0.377***	-0.211	0.397***	0.019	-0.004
	(0.106)	(0.218)	(0.018)	(0.047)	(0.003)
Post-Act 32	0.094***	0.177***	-0.132	0.006	-0.003***
	(0.019)	(0.027)	(0.090)	(0.006)	(0.001)
Pre-Act 32					
N	218,438	238,760	242,447	238,770	242,447
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.001	0.001	0.001	0.001
Dep. mean	3.675	5.845	6.696	0.093	0.943
Dep. SD	1.933	1.886	1.298	0.075	0.065
Post-Act 32					
N	218,036	238,336	242,033	238,346	242,033
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.001	0.001	0.001	0.001
Dep. mean	3.672	5.843	6.695	0.093	0.943
Dep. SD	1.932	1.885	1.297	0.075	0.065

Notes: Cluster-robust standard errors in parentheses. Treated cohorts split at first event year $c < 2012$

Table 19: Pre-/post-Act-32 split — school district level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind					
Pre-Act 32	0.027	0.057	0.106***	0.003	-0.000
	(0.038)	(0.046)	(0.030)	(0.004)	(0.004)
Inc-becomes					
Pre-Act 32	0.290*	0.155	0.272**	0.008	0.006
	(0.162)	(0.144)	(0.128)	(0.006)	(0.009)
Post-Act 32	0.043	0.523***	0.866***	-0.052	0.086
	(0.607)	(0.112)	(0.278)	(0.054)	(0.060)
Inc-nobind					
Pre-Act 32	0.794	1.149*	0.421***	-0.035	-0.015
	(0.506)	(0.633)	(0.148)	(0.023)	(0.021)
Dec-bind					
Post-Act 32	0.056	0.049	0.178***	-0.000	0.006***
	(0.075)	(0.111)	(0.044)	(0.004)	(0.001)
Dec-nobind					
Pre-Act 32	-0.126***	0.155***	0.076***	-0.032***	0.021***
	(0.025)	(0.028)	(0.016)	(0.002)	(0.003)
Pre-Act 32					
N	51,382	51,416	51,416	51,416	51,416
R^2	0.986	0.987	0.987	0.953	0.977
Within R^2	0.004	0.003	0.004	0.004	0.003
Dep. mean	6.261	8.335	8.769	0.125	0.900
Dep. SD	1.298	1.202	0.909	0.068	0.076
Post-Act 32					
N	50,947	50,981	50,981	50,981	50,981
R^2	0.986	0.987	0.987	0.953	0.977
Within R^2	0.003	0.001	0.003	0.004	0.005
Dep. mean	6.262	8.335	8.770	0.125	0.900
Dep. SD	1.297	1.203	0.908	0.068	0.076

Notes: Cluster-robust standard errors in parentheses. Treated cohorts split at first event year $c < 2012$ (Pre-Act 32) vs $c \geq 2012$ (Post-Act 32). Cells omitted (no estimable coefficients): Dec-becomes-non. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 20: Pre-/post-Act-32 split — block level

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc-bind				
Pre-Act 32	-0.218***	-0.118**	-0.037	0.037
	(0.059)	(0.058)	(0.037)	(0.037)
Post-Act 32	-0.116**	-0.022	-0.035***	0.035***
	(0.051)	(0.126)	(0.012)	(0.012)
Inc-becomes				
Pre-Act 32	-0.001	-0.041	-0.034***	0.034***
	(0.058)	(0.033)	(0.013)	(0.013)
Post-Act 32	-0.015	-0.062**	-0.053***	0.053***
	(0.026)	(0.028)	(0.014)	(0.014)
Inc-nobind				
Pre-Act 32	-0.114***	-0.132**	-0.033***	0.033***
	(0.033)	(0.066)	(0.008)	(0.008)
Post-Act 32	0.672***	0.106**	-0.147***	0.147***
	(0.086)	(0.052)	(0.036)	(0.036)
Dec-bind				
Pre-Act 32	0.368**	0.662***	-0.024	0.024
	(0.171)	(0.095)	(0.039)	(0.039)
Post-Act 32	0.045	-0.441***	-0.077***	0.077***
	(0.093)	(0.130)	(0.018)	(0.018)
Dec-becomes-non				
Pre-Act 32	-1.004***	0.804***	-0.151***	0.151***
	(0.155)	(0.185)	(0.042)	(0.042)
Post-Act 32	0.494***	0.364***	-0.073***	0.073***
	(0.067)	(0.065)	(0.028)	(0.028)
Dec-nobind				
Pre-Act 32	0.341***	-0.585**	0.006	-0.006
	(0.030)	(0.260)	(0.008)	(0.008)
Post-Act 32	-0.025	0.235***	0.006	-0.006
	(0.054)	(0.011)	(0.006)	(0.006)
Pre-Act 32				
N	10,977,594	4,091,601	10,977,594	10,977,594
R^2	0.852	0.864	0.589	0.589
Within R^2	0.000	0.000	0.000	0.000

All three estimating units use the stacked-continuous DiD with binding-state cells, re-estimated separately on the pre-2012 and post-2012 cohort subsamples. The municipality and SD splits use muni-by-stack and SD-by-stack fixed effects respectively, matching their pooled headlines; the block split uses block-by-stack and year-by-stack fixed effects, matching the block headline.

The binding-cell effect strengthens after enforcement consolidation. At the municipality level, the Inc-bind coefficient on the log count of residents working in their own municipality is -0.339^{**} in the pre-2012 sub-sample and -0.534^{***} in the post-2012 sub-sample—roughly a 50% larger response after Act 32 makes the EIT binding through consolidated employer withholding. The same pattern holds on the log count of in-commuters (-0.156 pre vs -0.334^{***} post), `log(employed residents)` (-0.098^{**} vs -0.104^{***}), and the share of workers living in their own muni (-0.047^{***} vs -0.040^{***}). The pre-Act-32 Inc-bind effects are non-zero and statistically detectable on most outcomes, so the binding-rule mechanism was not entirely inert before 2012; the enforcement break amplifies the response rather than activating it from a null baseline. The pooled headline (-0.401^{***} on the log count of residents working in their own municipality; Section 6.4) sits between the pre and post estimates, as expected.

The within-design placebo (Inc-nobind) is null on the headline outcomes in both periods (the log count of residents working in their own municipality -0.051 pre, -0.094 post; the share of workers living in their own muni -0.013 pre, -0.010 post). The institutional rule's predicted-null cell is satisfied regardless of the enforcement regime, which is consistent with the binding-rule interpretation: cells where the bill never moves should not respond in either period.

Block-level Inc-bind is roughly constant pre and post. At the block level, the muni-cohort Inc-bind coefficient on the log count of workers living in the block is comparable in the

two periods (the block-level effect predates the post-2012 enforcement break, contradicting a story in which all of the binding-channel response is driven by withholding consolidation). One reading: at fine geography, even pre-2012 statutory rate changes were salient enough through other channels (taxpayer reporting, local awareness) to induce sorting. The Act-32 amplification at the municipality scale may reflect aggregation rather than a discontinuous mechanism change.

Connection to Bagchi (2022). Bagchi (2022) documents that Act 32 increased PA EIT compliance via consolidated employer withholding. Our pre/post decomposition shows that the binding-cell residential response also strengthens after 2012 at the municipality scale; the two findings are complementary—compliance enforcement and behavioral response both grow under Act 32.

7.3 Dose-Magnitude Linearity

The stacked-continuous specification imposes linearity in dose: $\beta_k^s \cdot \Delta\tau_{it}^s$. Whether the per-pp coefficient is constant across small and large rate changes is an assumption rather than an output of the design. We test it by splitting Inc-bind units into terciles of total post-event dose at each estimating unit and re-estimating the headline regression for each tercile separately, with NC controls as in the headline.

Table 21: Dose-magnitude linearity — municipality level

Outcome	Log(work in own muni)	Log(employed residents)	Share live in muni
small	-0.448*	0.116	-0.038*
	(0.258)	(0.116)	(0.021)
medium	-0.114	0.004	-0.076
	(0.751)	(0.170)	(0.069)
large	-0.416***	-0.136***	-0.043***
	(0.091)	(0.024)	(0.010)

Notes: Cluster-robust standard errors in parentheses. Inc-bind units split by tercile of cumulative $\Delta\tau$ at +5. Per-pp dose-response should be approximately constant under linearity. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 22: Dose-magnitude linearity — school district level

Outcome	Log(work in own muni)	Log(employed residents)	Share live in muni
small	0.048	0.211***	0.000
	(0.119)	(0.071)	(0.009)
medium	0.133	0.277**	0.004
	(0.105)	(0.137)	(0.006)
large	0.059	0.094***	0.003
	(0.038)	(0.034)	(0.004)

Notes: Cluster-robust standard errors in parentheses. Inc-bind units split by tercile of cumulative $\Delta\tau$ at +5. Per-pp dose-response should be approximately constant under linearity. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 23: Dose-magnitude linearity — block level

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni
small	-0.229	-0.164	0.007
	(0.144)	(0.203)	(0.042)
medium	-0.210	-0.232***	-0.048**
	(0.165)	(0.021)	(0.021)
large	-0.176***	-0.133***	-0.065**
	(0.011)	(0.042)	(0.030)

Notes: Cluster-robust standard errors in parentheses. Inc-bind units split by tercile of cumulative $\Delta\tau$ at +5. Per-pp dose-response should be approximately constant under linearity. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 24: Dose-tercile robustness on Inc-bind, log(live_and_work)

Dose tercile	n munis	Coefficient	SE
Small	23	-0.440^{*}	0.259
Medium	4	-0.113	0.748
Large	13	-0.417^{***}	0.091
Pooled (headline)	40	-0.401^{***}	0.096

Notes: Inc-bind munis split by total cumulative $\Delta\tau^{\text{muni}}$ at year +5 post-event (or maximum observed). Small / medium / large terciles have dose ranges $[0.10, 0.20]$ / tied at 0.20 / $[0.30, 1.00]$ respectively; the medium tercile is thin ($n = 4$) due to mass ties at the 33rd percentile. Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The small and large dose terciles produce essentially identical per-pp coefficients (-0.440 vs -0.417). The medium tercile is too thin to identify but is direction-consistent. The linearity assumption is supported: the per-pp dose-response does not vary meaningfully across small and large rate hikes within the binding cell.

7.4 Alternative Control Group: Clean-in-Window

Our headline uses NC munis (no rate change anywhere in 2002–2019) as controls. **?** use a slightly relaxed definition: "clean controls" are units with no events in the window $[c - K, c + K]$ around each focal event, allowing units with events *outside* this window to serve as controls in cohorts whose window does not contain their event. The clean-in-window pool is a strict superset of NC.

Table 28 compares the headline coefficients on Inc-bind, Inc-becomes, and Inc-nobind cells across the two control definitions on six outcomes. The clean-in-window check is run at all three estimating units.

Table 25: Alternative control (clean-in-window) — municipality level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind	-0.321***	-0.032	-0.085**	-0.046***	0.036**
	(0.102)	(0.100)	(0.042)	(0.010)	(0.015)
Inc-becomes	-0.143*	-0.041	-0.081**	-0.018*	0.017**
	(0.075)	(0.082)	(0.034)	(0.010)	(0.007)
Inc-nobind	0.127	0.100	0.215**	-0.018*	0.014
	(0.173)	(0.195)	(0.085)	(0.009)	(0.013)
Dec-bind	0.041	-0.043	-0.047	0.040	-0.005
	(0.132)	(0.147)	(0.211)	(0.032)	(0.009)
Dec-becomes-non	-0.501	-0.083	-0.583	-0.087	0.028*
	(0.865)	(0.265)	(0.776)	(0.071)	(0.015)
Dec-nobind	0.260**	0.042	0.089	0.005	-0.006*
	(0.126)	(0.130)	(0.144)	(0.022)	(0.003)
N	—	—	—	—	—
R^2	—	—	—	—	—
Within R^2	—	—	—	—	—
Dep. mean	—	—	—	—	—
Dep. SD	—	—	—	—	—

Notes: Cluster-robust standard errors in parentheses. Control pool restricted to munis with no rate change anywhere in the event window. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 26: Alternative control (clean-in-window) — school district level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind	0.046	0.175***	0.132**	-0.016**	0.012
	(0.076)	(0.065)	(0.064)	(0.007)	(0.009)
Inc-becomes	1.732*	2.005**	1.136	-0.034	-0.005
	(1.027)	(0.946)	(0.729)	(0.026)	(0.032)
Dec-bind	0.114	0.029	0.148***	0.009	-0.001
	(0.110)	(0.113)	(0.051)	(0.007)	(0.008)
N	—	—	—	—	—
R^2	—	—	—	—	—
Within R^2	—	—	—	—	—
Dep. mean	—	—	—	—	—
Dep. SD	—	—	—	—	—

Notes: Cluster-robust standard errors in parentheses. Control pool restricted to SDs with no rate change anywhere in the event window. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind, Dec-becomes-non, Dec-nobind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 27: Alternative control (clean-in-window) — block level

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc-bind	-0.163***	0.084	-0.023	0.023
	(0.047)	(0.098)	(0.028)	(0.028)
Inc-becomes	-0.025	-0.003	-0.037***	0.037***
	(0.027)	(0.053)	(0.009)	(0.009)
Inc-nobind	0.028	-0.045	-0.022	0.022
	(0.081)	(0.079)	(0.021)	(0.021)
Dec-bind	0.060	0.387	-0.041	0.041
	(0.081)	(0.510)	(0.033)	(0.033)
Dec-becomes-non	0.544*	-0.336**	-0.046	0.046
	(0.319)	(0.169)	(0.063)	(0.063)
Dec-nobind	0.240***	-0.097	0.023***	-0.023***
	(0.085)	(0.183)	(0.007)	(0.007)
N	—	—	—	—
R^2	—	—	—	—
Within R^2	—	—	—	—
Dep. mean	—	—	—	—
Dep. SD	—	—	—	—

Notes: Cluster-robust standard errors in parentheses. Control pool restricted to blocks whose muni has no rate change in the event window. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 28: Alternative controls: NC vs clean-in-window

Outcome	Cell	NC-only (headline)	Clean-in-window
log(live_and_work)	Inc-bind	−0.401 (0.096)	−0.400 (0.096)
	Inc-becomes	−0.181 (0.072)	−0.181 (0.071)
	Inc-nobind	−0.062 (0.169)	−0.059 (0.167)
log(in_commuters)	Inc-bind	−0.213 (0.083)	−0.206 (0.082)
log(total_residents)	Inc-bind	−0.100 (0.033)	−0.100 (0.033)
share_live_in	Inc-bind	−0.045 (0.010)	−0.045 (0.009)
share_work_outside	Inc-bind	+0.041 (0.016)	+0.040 (0.016)

Notes: NC-only restricts controls to munis with no rate change anywhere in 2002–2019. Clean-in-window extends the control pool to munis whose first event falls outside the focal cohort's $[c - K, c + K]$ window. Clean-in-window adds 692 munis to the control pool relative to NC-only (1,945 vs 1,945 + 692 = 2,637 control-eligible munis). Cluster-robust SEs by muni in parentheses.

Coefficients are essentially identical across the two control definitions. The result is robust to the strictness of the control-group restriction.

8 Discussion

The continuous stacked DiD design produces four substantive findings. The block-level model is the headline causal estimate; the municipality-level model is the cross-check and the typical-jurisdiction magnitude calibration; the school-district-level aggregation is reported in the Companion Appendix for completeness.

Sorting is identified through the binding channel. The Inc-bind cell—rate hikes that move the bill—reduces the log count of workers living in the block at the block level by approximately 19 percent per pp (−0.21 log points, SE = 0.054). The municipality-level analogue is larger (−0.40 log points on the log count of residents working in their own municipality, ≈ 33% per pp). The Inc-nobind placebo—rate hikes whose post-event home

stack remains below the nonresident rate—is statistically indistinguishable from zero on the block-level count outcomes (-0.056 on the log count of workers living in the block, -0.113 on the log count of workers at the block) and on the municipality-level the log count of residents working in their own municipality (-0.057). Inc-becomes lies between the two. The pattern matches the institutional rule's prediction: the same statutory rate change can produce a strong, moderate, or null behavioral response depending solely on whether the rate moves the bill. This is direct evidence that workers respond to taxes they actually pay rather than to statutory rates per se.

Block-level identification has cleaner pre-trends than the municipality aggregation. The realized-dose Sun–Abraham event study at the municipality level shows a positive pre-event lead on the log count of residents working in their own municipality for the Inc-bind cell ($\beta_{\text{pre}} = +0.149$, $|t| = 2.66$), consistent with high-eventual-dose munis being on different trajectories before their rate change. At the block level the same diagnostic produces a near-zero pre-event coefficient on the log count of workers living in the block ($\beta_{\text{pre}} = -0.027$, $|t| < 1$), indicating that the gentrification-correlated pre-trend in the municipality coefficient is absorbed at finer geography. The block-level result is therefore the defensible per-pp causal quantity under the strong-parallel-trends assumption required by the recent continuous-DiD literature; the municipality-level estimate remains useful for back-of-envelope welfare calibration but should be cited with the pre-trend caveat.

Channel separation between muni and SD rate-setters. The muni resident-rate Inc-bind coefficient on resident outcomes is large and consistent at both the block and municipality levels. The SD-resident-rate cells, estimated jointly in the same regression, produce smaller and direction-mixed effects: $+0.177^{*}$ at the municipality level on the log count of residents working in their own municipality, $+0.097$ on block-level log count of workers living in the block ($p = 0.12$), and $+0.066^{*}$ at the SD level (sign-reversed). The SD rate is not the dominant driver of residential sorting through the binding channel. This is a substantive

change from an earlier draft that found a positive SD-rate "attraction" effect, which was an artifact of tract-centroid LODES aggregation; under block-level LODES with the unified two-rate-setter design, SD-rate effects on residents are small or wrong-signed.

Mechanism: local-services demand spillover. Industry-decomposed outcomes at the municipality level show consumer-facing jobs (retail, food, arts, other services) fall by approximately 26 percent in the Inc-bind cell (-0.30 log points, SE $= 0.15$) while tradeable jobs (manufacturing, wholesale, finance, professional services) are insulated (point estimate $+0.24$, statistically indistinguishable from zero). The composition shift is captured by the share of tradeable jobs rising approximately 11 percentage points in the binding cell. The decline in total jobs operates *via* local-services demand spillover from departing residents, not via direct nonresident-tax incidence.

Limitations. Three limitations merit acknowledgment. First, while the block-level stacked headline on the count outcomes is large, negative, and well-identified, the block-level share outcome (the share of block residents working in their own muni) is null in the binding cell (-0.035, $p = 0.26$) and marginally significant in the Inc-nobind placebo (-0.037, $p = 0.015$). The share outcome at block scale appears to reflect mechanical denominator dynamics on very small populations rather than binding-state-specific behavior, so the share-based block evidence should be read as suggestive rather than identified. The count-based block evidence remains the cleanest test. Second, the school-district aggregation produces direction-mixed coefficients and a marginally non-zero realized-dose pre-event lead; the SD-level results should be read as confirming the directional sign of the muni-rate effect under a coarser aggregation rather than as standalone identification of an SD-channel response. Third, the LODES OD data covers PA municipalities but not cross-state inflows in the intra-state main file used here; the auxiliary file (cross-state inflows) is downloaded but not yet incorporated, which would tighten the estimates of in-commuters for border munis.

9 Conclusion

This paper develops a continuous stacked difference-in-differences design for identifying the behavioral response to local income tax changes. The design exploits the piecewise structure of the Pennsylvania EIT ($T_{ji} = \max(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr,muni}})$) by classifying each rate-setter's events by direction and binding-state transition. The headline causal estimate is at the census block level, the only level at which both rate-setters' policy variation enters the regression at its native level; municipality-level results are reported as a coarser cross-check and as the typical-jurisdiction magnitude calibration; school-district-level aggregation results are in the Companion Appendix.

The headline finding is that a 1 pp muni resident-rate increase in the binding cell reduces the block-level count of workers living in the block (in logs) by approximately 19 percent (-0.21 log points, SE $= 0.054$) with a clean realized-dose parallel-trends test ($\beta_{\text{pre}} = -0.027$, $|t| < 1$). The municipality-level analogue is larger (-0.40 log points on the log count of residents working in their own municipality, $\approx 33\%$ per pp) but its pre-trend fails parallel trends ($\beta_{\text{pre}} = +0.149$, $|t| = 2.66$), consistent with selection of declining post-industrial cities into rate hikes; the block-level specification absorbs that pre-trend at finer geography. School district rate changes do not move residents through the binding channel at any estimating unit, even when SD changes flip the binding state. Industry decomposition at the municipality level shows the total-jobs decline is concentrated in consumer-facing sectors (retail, food, arts, other services), consistent with local-services demand spillover from departing residents rather than direct nonresident-tax incidence.

The design framework—a continuous stacked DiD spec estimated at the worker-resolution unit of observation, with two rate-setters' cells decomposed jointly by institutional binding state, a realized-dose event study as a strong-parallel-trends diagnostic, and a within-design placebo cell (Inc-nobind)—is portable to other settings where a piecewise tax rule generates predictable cross-sectional variation in whether a rate change moves the bill. Local sales taxes with use-tax-credit rules, county-level income-tax credits across state borders, and

special tax districts with overlapping jurisdictions are candidate extensions.

Acknowledgments

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A Pre-Treatment Balance

Table 29: Pre-Treatment Balance: Treated vs. Never-Treated PA Municipalities

Variable	Treated Mean	Control Mean	Norm. Diff.	N (T / C)
Population	14,174	3,711	0.139	1,265 / 39,688
Total Revenue ($)	25.7M	2.9M	0.077	1,267 / 39,692
EIT Revenue ($)	4.4M	438K	0.062	
RE Tax Revenue ($)	2.9M	538K	0.129	
Police Expenditure ($)	3.1M	424K	0.110	
Streets Expenditure ($)	1.5M	496K	0.121	
Total Debt ($)	28.6M	3.6M	0.089	
Market Value RE ($)	8.2M	2.1M	0.211	523 / 8,774
Total Index Crime	985	133	0.124	816 / 18,322
Building Permits	19.3	7.7	0.232	644 / 25,066
EIT Rate (%)	0.50	0.50	0.055	

Notes: Treated = municipalities that raised EIT during the sample period (pre-treatment years only). Control = municipalities that never raised EIT. Normalized differences < 0.15 for all fiscal variables except market value of real estate (0.211, limited pre-2012 coverage) and building permits (0.232, treated municipalities are larger and issue more permits in levels). Sources: PA DCED Annual Financial Reports, FBI UCR Offenses Known (93.5% agency match rate), Census Building Permits Survey (93.3% MCD match rate).

B Pre-Trend Tests on Fiscal Observables

We test the parallel trends assumption on eleven observables—six fiscal variables from the PA DCED Annual Financial Reports, real estate market values, total employment from LODES, Part I index crime from the FBI Uniform Crime Reports, and housing units permitted from the Census Building Permits Survey—using the Sun and Abraham (2021) estimator. For each outcome, we estimate Equation 6 with municipality and year fixed effects, clustering

standard errors at the municipality level. Table 30 reports the maximum absolute t-statistic among the pre-treatment coefficients at relative times $\ell \in \{-5, -4, -3, -2\}$.

Pre-trends are flat for nine of eleven observables: total revenue (max $|t| = 1.57$), EIT revenue (1.40), RE tax revenue (1.69), streets expenditure (0.70), population (1.09), total debt (1.36), total jobs from LODES (0.83), index crime (0.68), and building permits (1.36). None of the pre-treatment coefficients for these outcomes are individually significant at the 5% level. The jobs, crime, and permits results are particularly important: they confirm that treated and control municipalities were on parallel trajectories in employment, public safety, and housing development before the rate change, ruling out differential economic trends as confounds.

Police expenditure is an exception. The pre-treatment coefficients at $\ell = -5$ and $\ell = -4$ are negative and significant ($t = -2.85$ and $t = -2.42$, respectively), indicating that treated municipalities had relatively lower police spending 4–5 years before the rate increase. This pattern suggests that some EIT increases may have been associated with anticipated public-safety expansions financed by the new revenue. Importantly, the main results on the share of workers living in their own muni are robust to controlling for log police expenditure (the robustness section (Section 7)), confirming that this pre-trend does not confound the estimated sorting response.

Market value of real estate also shows a pre-trend at $\ell = -3$ ($t = -2.59$), but this variable has limited coverage (pre-2012 only, $N = 9{,}338$) and the pre-trend may reflect composition effects in the truncated sample rather than a genuine violation of parallel trends.

Table 30: Pre-Trend Tests: Sun-Abraham on Observables (PA, 2002–2019)

	Pre-Treatment Coefficients					
Outcome	$\ell = -5$	$\ell = -4$	$\ell = -3$	$\ell = -2$	Max $\lvert t \rvert$	N
log(Total Revenue)	−0.076	−0.051	−0.012	−0.052	1.57	43,170
	(0.049)	(0.056)	(0.038)	(0.032)		
log(EIT Revenue)	−0.025	−0.033	0.027	0.049	1.40	42,278
	(0.051)	(0.052)	(0.032)	(0.037)		
log(RE Tax Revenue)	−0.025	−0.060	−0.011	−0.015	1.69	41,683
	(0.044)	(0.036)	(0.029)	(0.017)		
log(Police Exp)	−0.085***	−0.068***	0.011	0.037	2.85	23,464
	(0.030)	(0.028)	(0.052)	(0.039)		
log(Streets Exp)	−0.001	0.003	−0.043	−0.029	0.70	43,105
	(0.059)	(0.075)	(0.053)	(0.059)		
log(Population)	0.011	0.007	0.002	0.003	1.09	43,150
	(0.012)	(0.008)	(0.006)	(0.003)		
log(Total Debt)	−0.029	−0.051	−0.179	−0.103	1.36	24,863
	(0.176)	(0.214)	(0.132)	(0.102)		
log(Market Val RE)			−0.403***	−0.026	2.59	9,338
			(0.156)	(0.016)		
log(Total Jobs)	0.006	0.018	−0.001	−0.005	0.83	24,557
	(0.023)	(0.022)	(0.015)	(0.013)		
log(Index Crime)	−0.125	−0.094	−0.084	−0.031	0.65	20,613
	(0.194)	(0.151)	(0.236)	(0.155)		
log(Bldg Permits)	−0.030	−0.057	0.084	0.171	1.35	27,515
	(0.222)	(0.157)	(0.147)	(0.126)		

Notes: Each row reports Sun and Abraham (2021) estimates of the pre-treatment coefficients $\hat{\mu}_\ell$ for $\ell \in \{-5, -4, -3, -2\}$, with $\ell = -1$ as the reference period. Standard errors (in parentheses) are clustered at the municipality level. Max $|t|$ is the largest absolute t-statistic among the pre-treatment coefficients. Pre-trends are flat for all outcomes except police expenditure (lower spending at $\ell = -5$ and $\ell = -4$) and market value of real estate (limited to pre-2012 data, $N = 9{,}338$). Total jobs from LODES; index crime from FBI UCR Offenses Known (93.5% agency-to-municipality match);